

BeneFusion VP5

(This manual is also applicable to BeneFusion VP5 Ex Infusion Pump)

Infusion Pump

Maintenance Manual

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SK Medical is responsible for the effects on safety, reliability and performance of this product, only if:

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- The electrical installation of the relevant room complies with the applicable national and local requirements.
- The product is used in accordance with the instructions for use.

Preface

Manual Purpose

This Service Manual provides detailed introductions to the hardware composition, installation, dismantling, testing and troubleshooting of this product and its related accessories, so that the maintenance personnel can handle the common problems effectively. It does not provide in-depth information on the product structure and design principle. If you experience problems that cannot be solved, please contact our After-sale Service Department.

The introductions to this product in this Service Manual are based on the most complete configuration, so some of them may not be applicable to the product you're maintaining. In case of any question, please contact our after-sale service department.

Before product maintenance, please read this Service Manual carefully, and ensure that you have fully understood the content, so that you can maintain the product correctly, avoiding product damage or personal injury.

Intended Audience

This Service Manual is applicable to the professional biomedical engineers responsible for the product maintenance, the authorized maintenance personnel or the after-sale service representatives.

ConventionsVersion Information

The version number of this Service Manual may be updated without notice at any time due to change in the software or technical specification. The version information of this Service Manual is as follows:

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1 Safety

1.1 Safety Information

The safety statements presented in this chapter refer to basic safety information that the operator must pay attention to and abide by when using the infusion pump. There are additional safety statements in other chapters or sections, which may be the same as or similar to the following, or specific to particular operations.

Dangers

- Indicates an imminent hazard that, if not avoided, could result in death, serious injury or damage to product/property.
-

WARNING

- Indicates a potential hazard or unsafe practice that, if not avoided, could result in death, serious injury or damage to product/property.
-

CAUTION

- Indicates a potential hazard or unsafe practice that, if not avoided, could result in minor personal injury, product malfunction or damage to product/property.
-

NOTE

- Provides application tips or other useful information to ensure that you get the most out of the product.
-

1.1.1 Dangers

This Manual does not contain any information at the "Danger" level.

1.1.2 WARNING



-
- **Device, cables and accessories must be inspected before use to guarantee their normal and safe operation.**
 - **This equipment can only be connected to the socket with ground protection. Please adopt a rechargeable battery instead of the socket as the power supply if the socket is not provided with a ground lead.**
 - **To prevent fire or explosion, do not operate the system in the presence of anesthetic, flammable or explosive materials.**
 - **The patient's clinical condition and the working condition of the infusion pump must be monitored carefully, and the alarm volume and alarm levels need to be set according to the actual needs. Operation and performance relying solely on the auditory alarm system alone is not sufficient, and setting the alarm at a low volume may endanger the patient. If the alarm volume is less than the surroundings volume, which can further lead to operators identify alarm mistakenly.**
 - **Please carefully install the power line and cables with various accessories to prevent the patient from choking or suffocation caused by entanglement of the cables or by electrical disturbance.**
 - **The packaging materials must be disposed of in compliance with local laws and regulations or the hospital policy on waste management. They must be kept out of the reach of children.**
 - **This equipment must be used with professional medical consumables, and its accuracy cannot be guaranteed when it is used with an infusion set that is a non-standard consumable or a consumable without calibration.**
-

1.1.3 CAUTION



- Use the accessories specified in this Operator's Manual to guarantee the patient's safety.
 - Electromagnetic fields may affect equipment performance. This makes it necessary for other equipment used in the vicinity of the pump to meet EMC standards. Mobile phones, X ray and MRI equipment are all potential interference sources because of their high-intensity electromagnetic radiation.
 - Before the equipment is connected to the power supply, check that the voltage and frequency of the power supply match the specifications on the label or in this Operator's Manual.
 - Please install and carry the equipment correctly to protect the equipment from damage from drops, impacts, violent shaking or other external mechanical forces.
 - Avoid direct sunshine, high temperatures and dampness.
 - Check the built-in battery before use to make sure it has sufficient power. Recharge the battery if necessary.
-

1.1.4 NOTE

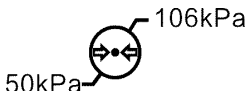
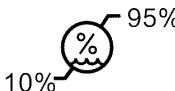
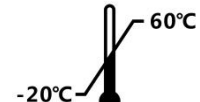
NOTE

- The software of the equipment is developed according to the software development demands of IEC60601-1 standard, which can minimize the possibility of the risk caused by program error.
 - This Operator's Manual describes the most complete functional configuration of the equipment. The product you are using may not have some of the settings or functions described herein.
 - Do not insert devices that are not specified by the manufacturer into the multifunction interfaces.
-

1.2 Equipment Symbols

The equipment you purchased may not provide you with all the following symbols.

	NOTE ! Refer to the accompanying document (This Manual)		ON/OFF
	Alternating current power supply (AC)		Battery
	Alarms		Silence
	Clear/Back		Start
	Bolus		Confirm
	Stop		Menu
	Move up/Increase		Move down/Decrease
	Move left		Move right
	Infusion set		Selected drug
	Night mode		Equipotential port
	Drop sensor interface		Multifunction interface
	Wireless networking, normal communication with BeneFusion CS5 Infusion Supervision System		Wireless networking, no communication with BeneFusion CS5 Infusion Supervision System
	Wireless network is establishing a network connection		Current pump communicated successfully with BeneFusion DS5 Infusion Supervision System
	Current pump has downward relayed No.X pump		Current pump has forward relayed No.X pump
	Executed prescription		Pause prescription

	Executing prescription		Non-executed prescription
	Caution		Lock
IP23	Protected against solid foreign objects with a diameter no less than 12.5mm and protected against spraying liquid water		Protected against defibrillation CF applied parts
	Date of manufacture		Manufacturer
	Electronic equipment: dispose of separately to avoid polluting the environment		Wireless transceiver
	The European Union Representative Office		Serial number
	Environmentally-friendly use periods of electronic products (20 years)		CE mark
	Package should be kept between 50–106 kPa during transport		Package should be kept between 10%–95% humidity during transport
	Package should be kept between -20–60°C during transport		

2 Design Principle

2.1 Product Overview

The Infusion Pump is used in conjunction with the infusion set to control the dose of liquid infused into the patient's body.

The Infusion Pump is suitable for adults, children and newborns in clinical departments.

The Infusion Pump is expected to be used in institutes or units with healthcare capabilities. This includes but is not limited to: outpatient departments, emergency departments, wards, ICU, operating rooms, observation rooms, clinics, and nursing homes.

The Infusion Pump primarily consists of a housing, a motor drive system, an input system, a memory system, a control system, a display system, a sensor-based monitor system and an alarm system. Since some parts and functions are optional, the Infusion Pump you purchased may not contain these additional parts and their relevant functions.

This Infusion Pump has the following functions and characteristics:

- Accurate control of flow rate, volume and infusion time; real-time monitoring of speed and direction of a running stepper motor; prevention of over-current, under-current and inverse-current.
- Good linear flow.
- Flow rate may be changed during infusion.
- History records: Operation, infusion and alarm logs can be recorded, and data can be exported.
- Drug library: Operation interface can display the name of the drug currently being infused.
- Data storage: At least 2000 history records and 2000 drug names can be stored.
- Memory function: Last infusion parameters can be memorized.
- Bolus functions: Manual bolus and auto bolus.
- Standby function: Parameters prior to standby can be retained.
- Purge function: Air bubbles in the infusion set are eliminated before the infusion.
- Dynamic pressure monitoring function: Providing real-time testing of infusion set pressure and with a pressure icon, indicating pressure status (i.e. normal pressure, near occlusion alarm pressure and occlusion alarm pressure status).
- Automatic and manual lock-screen functions: Preventing accidental operation

and misuse.

- Anti-bolus function: Automatically releasing line pressure after occlusion.
- Self-test on startup function.
- Anti-defibrillation function.
- KVO function: Keeping the patient's line open.
- Super bright large LED display.
- Alarm: Light, sound, prompt and other alarm methods, alarm volume is adjustable.
- Network communication: Wireless communication with BeneFusion CS5 Infusion Supervision System and BeneFusion DS5 Infusion Supervision System, data can be exported and on-line upgraded via PC connection.
- Various infusion modes available.

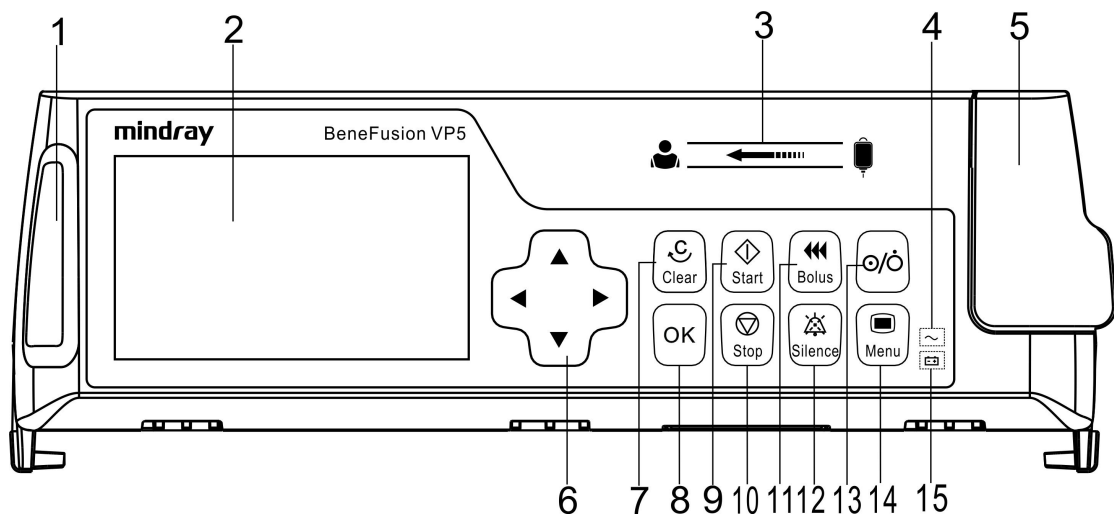
2.1.1 Structural Composition and Performance

The Infusion Pump primarily consists of a housing, a motor drive system, an input system, a memory system, a control system, a display system, a sensor-based monitor system and an alarm system. The wireless module is optional. Optional functions of the software comprise Rate Mode, Time Mode, Body Weight Mode, Ramp Up/Down Mode, Sequential Mode, Loading Dose Mode, Micro-infusion Mode, Drug Library, Record and Anti-bolus Function.

Since some parts and functions are optional, the Infusion Pump you purchased may not contain these additional parts and their relevant functions.

2.2 Host

2.2.1 Front View



1. Alarm light

The alarm light indicates different alarm levels in different colors and flash frequencies.

2. Display

Used for displaying infusion parameters and relevant content.

3. Liquid flow direction indicator

4. AC indicator light

- On: The pump is connected to an AC power supply.
- Off: The pump is not connected to an AC power supply.

5. Door holder

Pull it to open the door.

6. <DIRECTION>

Used for adjusting value, change lines and pages.

7. <CLEAR/BACK>

- Under non-setting status, indicate to returns to previous menu or operation.
- Under the setting status, indicate to clear the current set or cancel the edit.

8. <OK>

Used for confirming input operation and saving values.

9. **<START>**

After installing the infusion set correctly and completing setting infusion parameters, press this key to start the infusion.

10. **<STOP>**

During infusion, press this key to stop infusion. Infusion stops caused by alarms, such as occlusion and so on, press this key to cancel the alarm.

11. **<BOLUS>**

- During infusion, press this key to enter the Bolus Settings screen.
- When the pump is stopped, press this key to enter the purge settings screen.

12. **<SILENCE>**

- For high level and mid-level alarms, press this key to silence for 2 minutes. The alarm will be cancelled automatically if a new alarm is triggered within 2 minutes.
- For low-level alarms, press this key to cancel alarm.

13. **<POWER>**

- Used for turning power on, entering in standby state and turning off operations.
- When power off, press and hold (>3 s) the key.

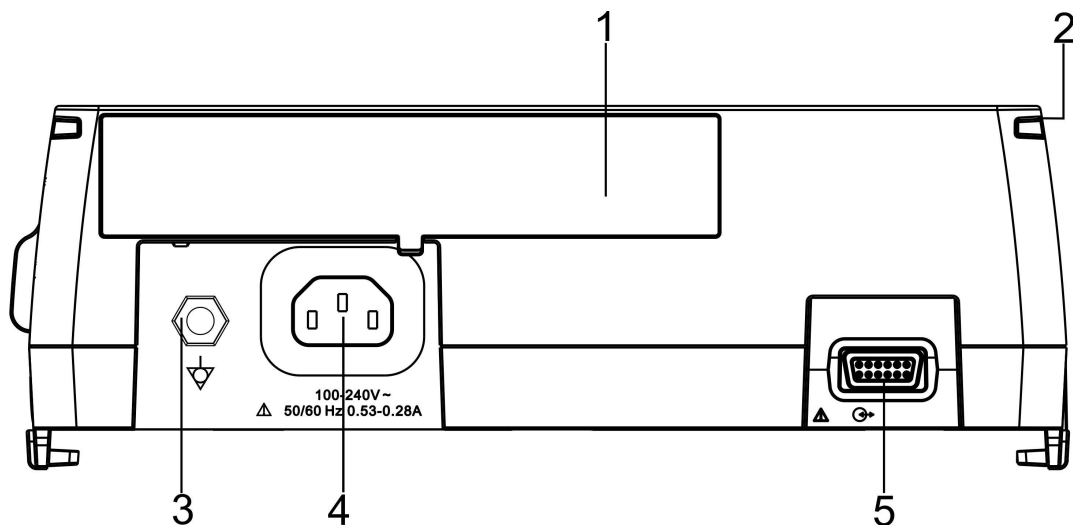
14. **<MENU>**

- Under non-operation status, used for switching [**Main Menu**] interface and other interfaces.
- Under operation status, press and hold this key to lock; in locked state, press and hold to unlock.

15. Battery indicator light

- Steady green indicates that the battery is charging.
- Flashing indicates that the battery is providing power.
- Light off indicates that there is no battery or the equipment is turned off and not connected to an AC power supply.

2.2.2 Rear View



1. Battery box

2. Multi-channel pumps connect rail

Used for combining multi-channel pumps.

3. Equipotential port

When other device and infusion pump are used together, please adopt wires to connect other devices and equipotential port of the pump, to eliminate ground potential difference between different devices and ensure safety.

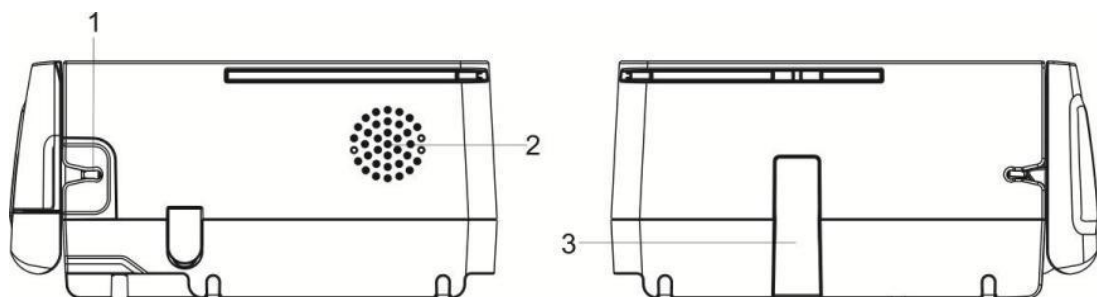
4. Alternating current power supply (AC) port

Connected by three-core-type power cord and AC power source.

5. Multifunction interface, which combines the following interface functions:

- DC power input interface
- RS232 interface
- Nurse call interface

2.2.3 Side View



1. Tube seat

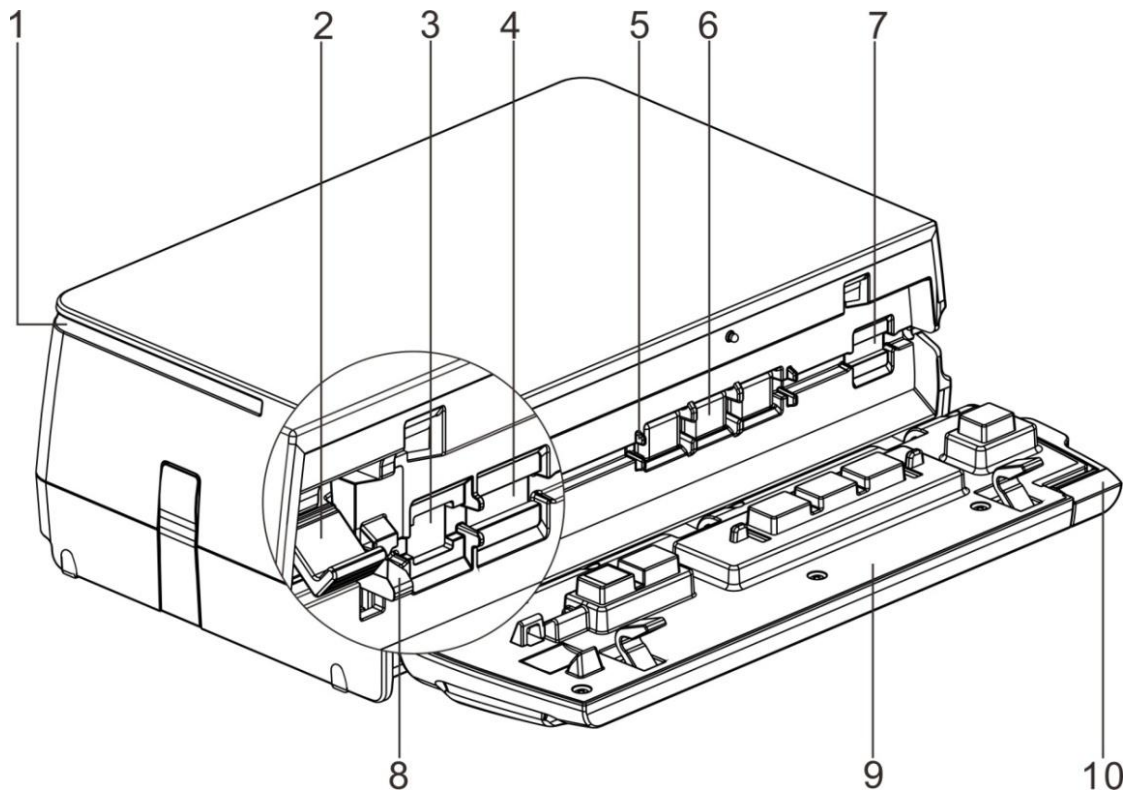
2. Speaker hole

3. Multi-channel pumps hasp

The multi-channel pumps are connected to secure the equipment.

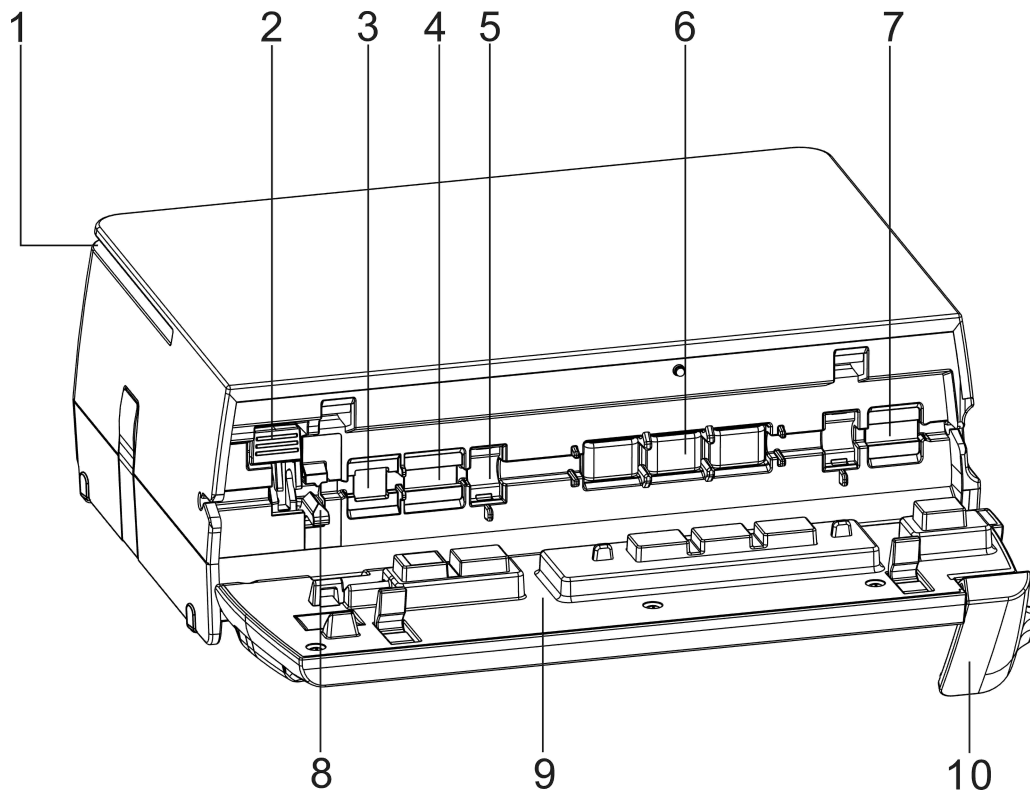
2.2.4 Side View with the Door Opened

2.2.4.1 Infusion Pump with Conventional Infusion Sets



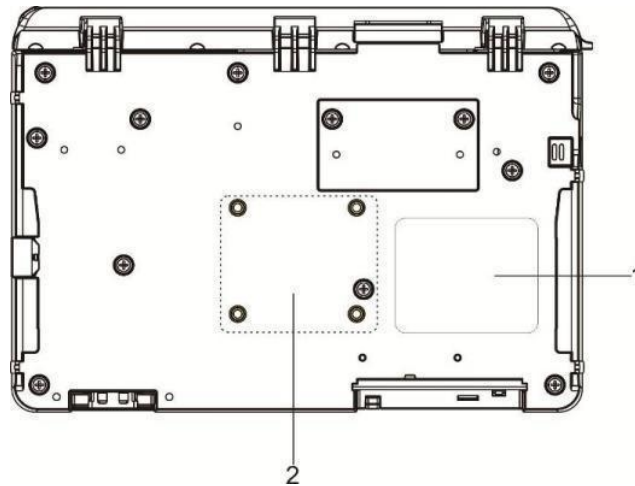
1. Connection trough for multiple pumps
2. Liquid check clip button
3. Bubble checkout slot
4. Downward pressure checkout slot
5. Infusion set slot
6. Waterproof membrane
7. Upward pressure checkout slot
8. Liquid check rod
9. Door
10. Door holder

2.2.4.2 Infusion Pump with Specified Infusion Sets



1. Connection trough for multiple pumps
2. Liquid check clip operating lever
3. Bubble checkout slot
4. Downward pressure checkout slot
5. Locating slot
6. Waterproof membrane
7. Upward pressure checkout slot
8. Liquid check rod
9. Door
10. Door holder

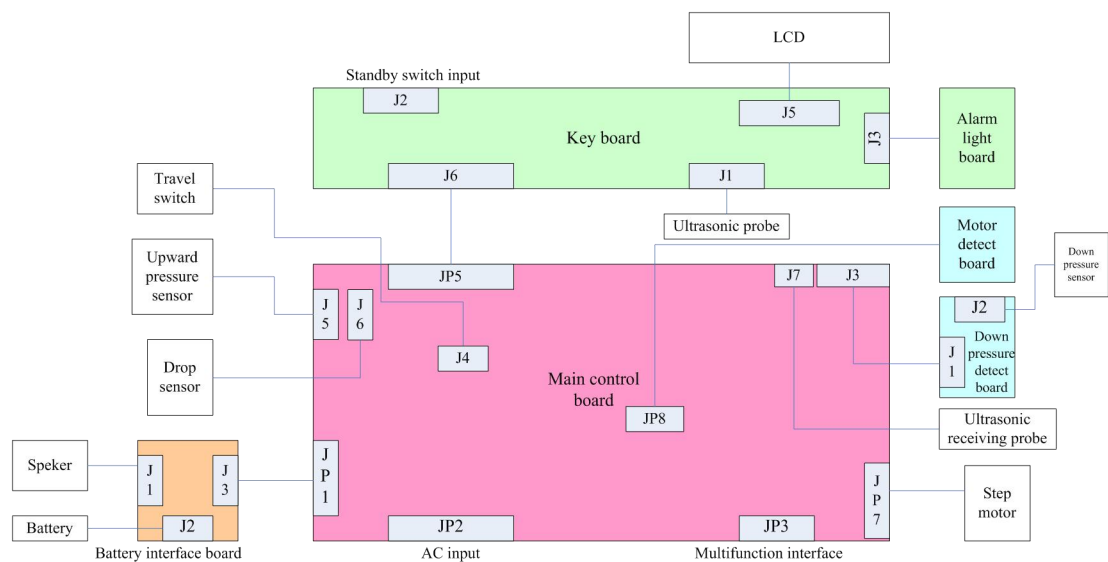
2.2.5 Bottom View



1. Product label
2. Fixed mounting holes

2.3 Hardware Principle

Hardware board framework of BeneFusion VP5 Infusion Pump is shown in the following figure, which includes main control board, keyboard and downward pressure detection board.



2.3.1 Main Control Board

VP5 Main Control Board consists of three sections, namely Power, Control and Monitoring sections.

Power section includes AC/DC conversion module, switch control circuit, DC power conversion circuit, power management CPU, standby sound and light alarm circuit, battery charge and discharge management circuit.

The AC/DC conversion module converts AC 100-240VAC input, 50/60Hz to DC 15V output.

Switch control circuit combines inputs from AC/DC, external DC and internal battery into VPC, with AC/DC and external DC having higher priority than internal battery; switch signal then converts VPC to VPP as output.

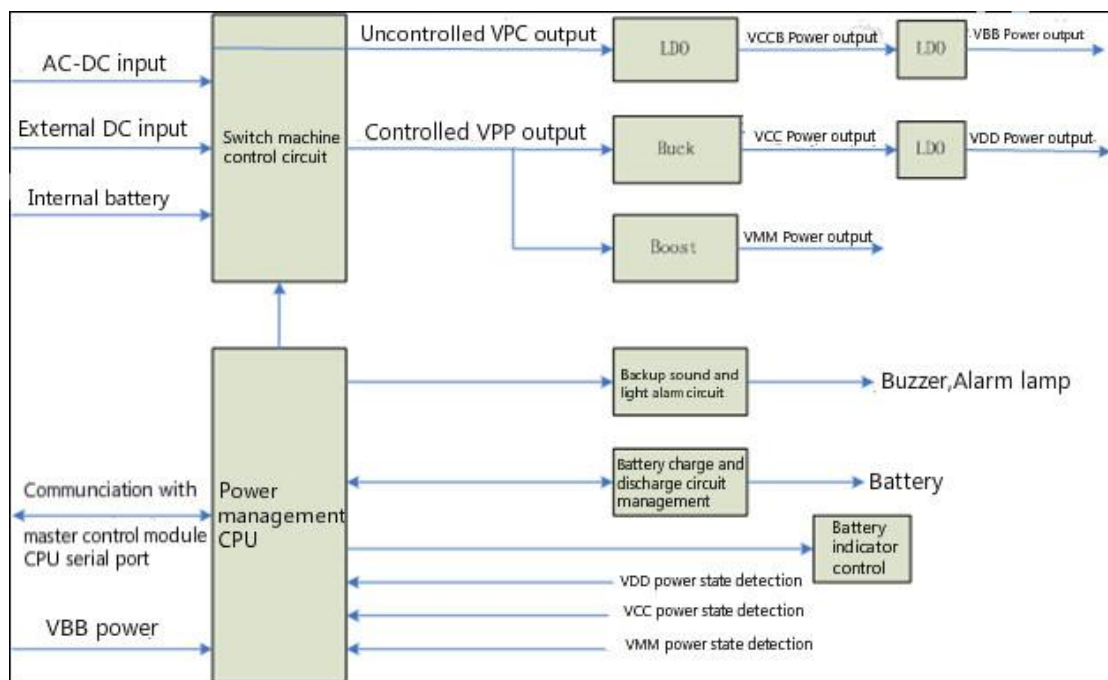
While VPC is not controlled by switch signal, VPP output is controlled by switch signal. Via LDO circuit (U7 ZSR500GTA) VPC is bucked to VCCB (5V) and via LDO circuit (U3 SPX5205) VCCB is bucked to VBB (3.3V); Via Buck circuit (U12 TPS5410D) VPP is bucked to VCC (5V); via LDO circuit (U13 SPX1117M3) VCC is bucked to VDD (3.3V); via Boost circuit (U26 LT3757EMSE) VPP is boosted to VMM (+24V).

Power management CPU is responsible for switch control, motor power-down protection under a single fault, battery charge status monitoring, various module power status detection, battery indicator control, and communication with the main control module CPU via serial ports. Power management CPU uses Cortex M0 core chip LPC1112 (U10) powered by VBB power supply. Once powered off, the CPU is in sleeping mode. After power switch on the keyboard is pressed, triggering wake-up mode, power-on and module power in-use signals, which leads to output of VCC, VDD and VMM in this order and the entire system is then powered on. Main control CPU is powered by VDD power supply. After the CPU is powered on, it outputs a boot lock signal, preventing accidental shutdown of power management CPU. In case of a normal shutdown, power management CPU first sends a request via serial ports to main control CPU which then revokes the boot lock signal after confirming that shutdown is permitted. Finally shutdown is completed by the power management CPU and the system then goes to sleeping mode again.

Battery charge and discharge management circuit uses BQ24103RHLLR, with off-mode charging current designed for $1300 \pm 100\text{mA}$ (charging time is not more than 6 hours) and on-mode charging current designed for $650 \pm 50\text{mA}$ (charging time is not more than 12 hours); the battery is designed to be fully charged at a voltage of $8.4 \pm 0.2\text{V}$.

Under the circumstance of the battery voltage being too low and the AC power being switched off, the standby sound and light alarm circuit triggers a buzzer which rings at a frequency of $2.73\text{Hz} \pm 0.5\text{V}$ and a red alarm LED is continuously lit.

When the power management CPU detects unusual power of the VDD or abnormal communication with the main control module CPU, the buzzer rings and the stepper motor's power is then shut down.



Key signal testing points:

Tested by a multimeter D2 positive ground voltage (ACDC_IN) is about 14V-16V.

Tested by a multimeter D3 positive ground voltage (EDC_IN) is about 10V-16V.

Tested by a multimeter D4 terminal voltage (BAT_IN) is about 6.8V-8.4V.

Tested by a multimeter TP65 (VPC) is about 6.4V-15.6V.

Tested by a multimeter TP66 (VPP) is about 6.4V-15.6V.

Tested by a multimeter TP60 (VCCB) is about 4.5V-5.2V.

Tested by a multimeter TP61 (VBB) is about 3.0V-3.6V.

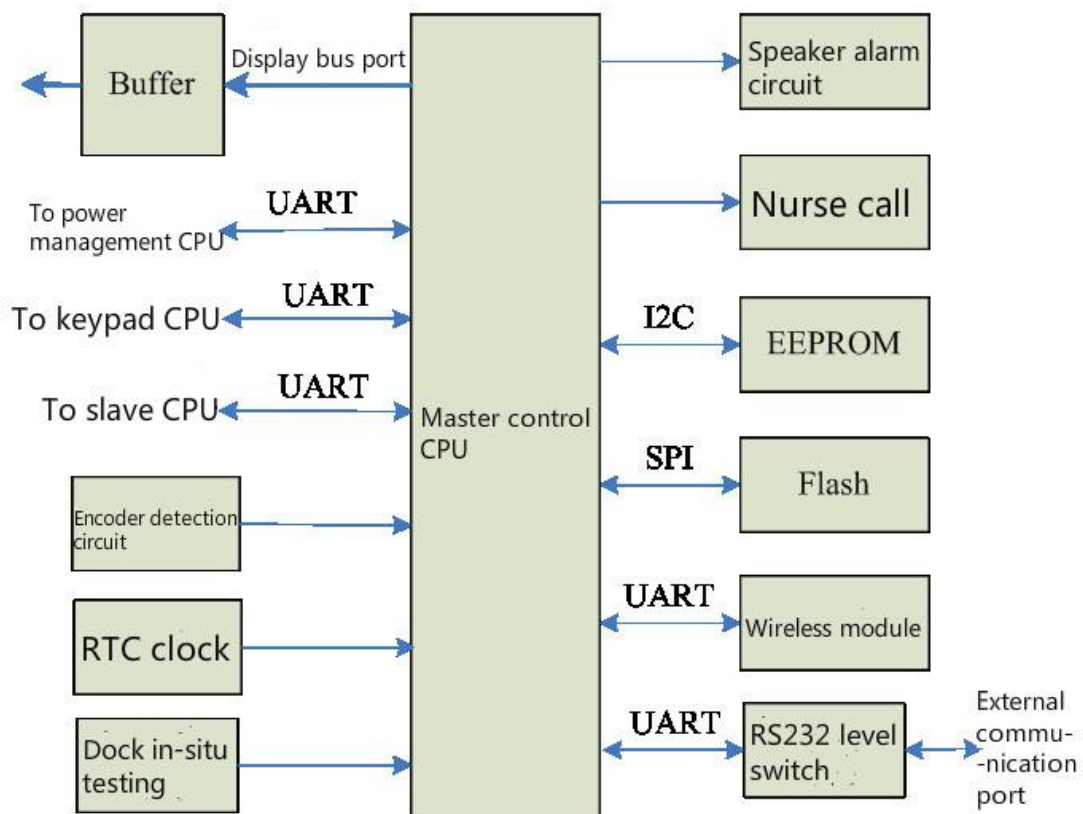
Tested by a multimeter TP62 (VCC) is about 4.5V-5.5V.

Tested by a multimeter TP1 (VDD) is about 3.0V-3.6V.

Tested by a multimeter TP16 (VMM) is about 19V-22V.

Control section is responsible for LCD display driver, speaker alarm circuit, nurse call, Dock on-line detection and power CPU, key CPU, and adjunct CPU UART communication. Main control CPU uses Cortex M3 core chip STM32F103VE (U14), with Intel 8080 system bus interface as the display driver interface and Buffer chip U8 and U9 (SN74LVCR2245) protecting the main control CPU pin from being burned by external overshoot. Speaker alarm circuit consists of PWM wave

modulation circuit and amplifier circuit; Q5, U19 and peripheral RC form the PWM wave modulation circuit; U18 is a voice amplifier chip and power supply is +5V. Nurse call is an extended function, which is connected to the hospital alarm system via a designated nurse call cable; Q4, Q6 and peripheral RC form the nurse call circuit. U16 is EEPROM with an I2C interface for storing information such as historical records and calibration data. U15 is Nor flash with a SPI interface for storing drug libraries, language libraries and picture resources etc. U28 is a Zigbee module connected to the main control CPU via UART, enabling external wireless communication. X2 is a 32.768KHz passive crystal oscillator, providing the RTC module of the main control CPU with a clock. Encoder detection circuit is used for real-time detection of the motor's speed by the master control block CPU and whether its direction is consistent with the setting (adjunct CPU also detects in real-time the motor's speed and direction). U6 is a RS232 level converter chip; external communication interface, nurse call signals and Dock on-line detection signals are all connected to the multi-functional interfaces JP3. Main control CPU communicates with power management CPU, key CPU, and adjunct CPU via UART.



Key signal testing points:

RP5-6 foot LCD_WR writes control signals.

RP5-8 foot LCD_RS data / command selection signals.

RP6-2 foot LCD_CS chip-select signals.

RP6-6 foot UART signals sent by the main control CPU to the key CPU.

RP6-8 foot UART signals received by the main control CPU from the key CPU.

TP24 external serial UART signals sent by the main control CPU.

TP23 external serial UART signals received by the main control CPU.

TP40 UART signals received by the main control CPU from the power management CPU.

TP41 UART signals sent by the main control CPU to the power management CPU.

TP53 SDA signals with which the main control CPU communicates with EEPROM.

TP63 SCL signals with which the main control CPU communicates with EEPROM.

TP54 CS_F signals with which the main control CPU communicates with Flash.

R198 SCK_F signals with which the main control CPU communicates with Flash.

TP56 MISO_F signals with which the main control CPU communicates with Flash.

R197 MOSI_F signals with which the main control CPU communicates with Flash.

TP59 main control CPU reset signals.

TP5 UART signals received by the main control CPU from the adjunct CPU.

TP7 UART signals sent by the main control CPU to the adjunct CPU.

TP69 Dock on-line detection signals, low level valid.

TP64 CALL control signals, high level valid.

TP27 WIFI module UART signals received by the main control CPU.

TP28 WIFI module UART signals sent by the main control CPU.

TP26 WIFI module reset signals.

TP46 WAVE speaker control signals, 20KHZ square wave signals when the speaker sounds.

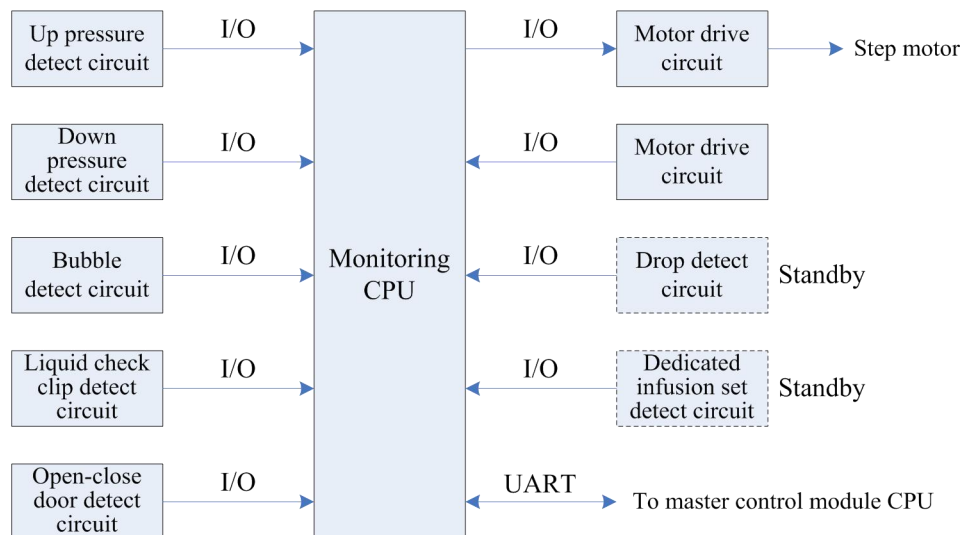
TP47 SAMPLE speaker control signals, 600HZ square wave signals when the speaker sounds.

TP49 voice chip in-use signals and high level valid.

TP21 motor detection of input signals of optocoupler 1, output pulse signals when the motor is running.

TP25 motor detection of input signals of optocoupler 2, output pulse signals when the motor is running.

Monitoring module consists of adjunct CPU, stepper motor drive circuit, motor detection circuit, upward pressure detection circuit, downward pressure detection circuit, bubble detection circuit, liquid clip detection circuit, door detection circuit, drop detection circuit and designated infusion set detection circuit. Adjunct CPU uses STM32F100C8 (U1); stepper motor drive circuit uses motor control chip TMC260-PA (U2). Motor detection circuit uses dual optocoupler to determine the motor's direction by comparing the two pulse sequences and to determine the motor's speed by measuring the frequencies of the two pulse sequences.



Liquid check clip detection circuit, dedicated infusion set detection circuit and motor detection circuit all use optocoupler. Upward pressure detection circuit and downward pressure detection circuit are located at upward pressure plate and downward pressure plate respectively; bubble detection circuit, dedicated infusion set detection circuit and liquid check clip detection circuit are located at bubble detection plate; door detection circuit uses a limit switch and motor detection circuit is located at motor detection plate.

Key signal testing points:

TP4 adjunct CPU reset signals, low level valid.

TP6 motor stall detection signals, high level valid.

TP13 S_NCS signals with which adjunct CPU communicates with U2.

TP12 S_SCK signals with which adjunct CPU communicates with U2.

TP11 S_SDI signals with which adjunct CPU communicates with U2.

TP14 S_SDO signals with which adjunct CPU communicates with U2.

TP10 enable signals with which adjunct CPU controls U2, low level valid.

TP9 signals with which adjunct CPU controls U2 direction, low level motor in forward rotation and high level motor in reversal rotation.

TP8 signals with which adjunct CPU controls U2 stepping, pulse signals at motor start.

MSD1 motor detection of input signals of optocoupler 1, output pulse signals when the motor is running.

MSD2 motor detection of input signals of optocoupler 2, output pulse signals when the motor is running.

TP105 AD signals for upward pressure detection.

TP34 AD signals for downward pressure detection.

TP20 signals for liquid check clip detection, low level when the clip is closed and high level when open.

TP19 signals for dedicated infusion set detection, high level when the tube is on-line and low level when off-line.

RP5-4 foot enable signals for ultrasonic sensor transmission, low level valid.

TP112 AD signals for bubble detection, 0.4-1.7V when the circuit is filled with liquid, less than 0.3V when the circuit is filled with bubbles.

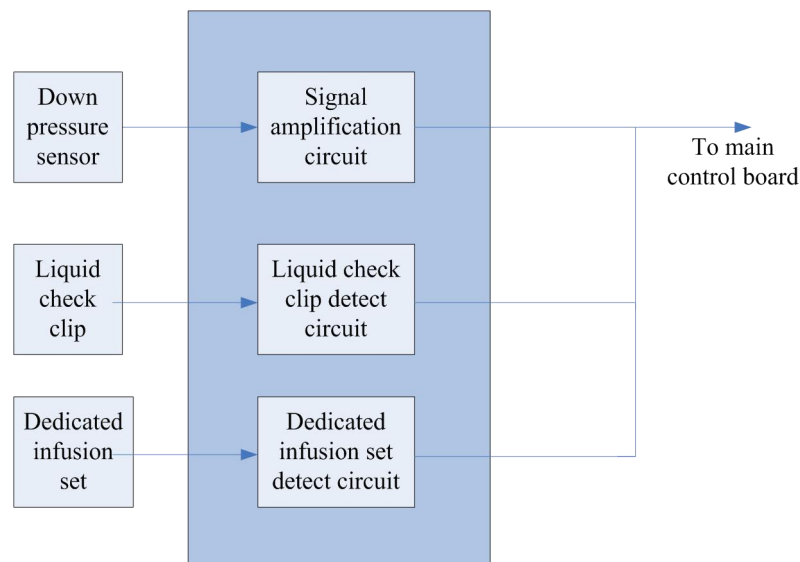
TP18 signals for door detection, high level when the door is open and low level when closed.

TP100 Tested by a multimeter, 5V power supply for drop sensor is about 4.5-5.5V.

TP101 enable signals for drop detection.

TP102 signals for drop detection.

2.3.2 Downward Pressure Detection Board



Weak signal output by pressure sensor is amplified by LMV824MX (U3) of downward pressure detect board, analog signal is sent to adjunct CPU to adjust the pressure. Liquid check clip is detected by optocoupler U1 and surrounding circuit, dedicated infusion set is detected by optocoupler U2 and surrounding circuit.

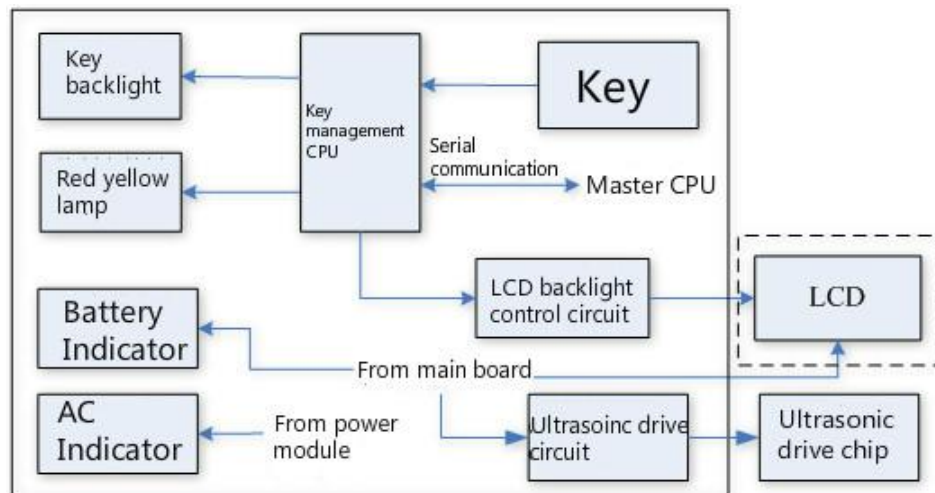
Key signal testing points:

TP1 3.3V for downward pressure detection, 3-3.6V tested by a multimeter.

J1-8 Analog signal output by downward pressure detection.

TP6 Zero adjustment PWM signal for downward pressure detection.

2.3.3 Keyboard



Key management CPU uses Cortex M0 core chip LPC1112 (U4) responsible for functions such as key (a total of 12 keys, of which the power switch key is directly linked to the power module CPU and other keys are processed by the key management CPU) scanning process, alarm light drive, and LCD backlight control (8 adjustable levels). A white key backlight is placed near each of the keys; once a key is pressed, the key backlight will be lit by the key management CPU. Alarm lights are in red and yellow colors and the flash frequency of each color is independently adjusted by software.

Battery indicator light is continuously lit during charging, flashes when powered by battery alone with the flash frequency being controlled by the power management CPU; AC indicator is continuously lit when AC power supply is normal and is off when AC is disconnected.

LCD control signals use Intel 8080 system bus interface, all from the main CPU.

Transmission drive circuit of the bubble detection ultrasonic sensor consists of U3 and peripheral circuits; control in-use signals are sent by the adjunct CPU and drive signals are connected to the ultrasonic transmitter chip via J1.

Key testing points:

Tested by a multimeter TP10 (VDD) is about 3.0V-3.6V.

Tested by a multimeter TP11 (VCC) is about 4.5V-5.5V.

R20 BL_PWM signals for LCD backlight control, PWM signals or high level switching on backlight.

Tested by a multimeter TP2 LCD backlight power supply is about 18.6V-20.4V.

TP20 signals for battery indicator light.

TP21 signals for AC indicator light.

TP15 enable signals for ultrasonic sensor drive, low level valid.

TP5 signals for ultrasonic sensor drive US_DRV1

TP6 signals for ultrasonic sensor drive US_DRV2.

3 Test and Maintenance

3.1 Description

To ensure long-term stability of the infusion pump, maintenance personnel must carry out regular inspection, maintenance and testing. This chapter provides the basic testing methods for the infusion pump, and recommends the proper testing frequency and testing tools. Maintenance personnel should choose proper testing tools to carry out inspection and testing according to the actual needs.

The tests and testing methods provided in this chapter are mainly used to verify whether the performance of the infusion pump can meet the specifications. If test results do not meet the specifications, it indicates that a functional module has failed, which must be repaired or replaced immediately. In case of any other questions, please contact our After-sale Service Department in a timely manner.

CAUTION

- All the tests must be performed by qualified maintenance personnel only.
- Prior to test, maintenance personnel shall ensure the applicability of the testing tools and connecting cables, and familiarize themselves with the use of these tools.

3.1.1 Test Report

After the professional maintenance personnel have performed the tests, please record in the following test report.

Test equipment			
Name		Equipment Model	Equipment No.
Test record			
Serial No.	Test content	Testing point	Test result
1			
2			
3			
4			
5			
Test judgment			
Pass or Fail:		Tester:	Test date:

3.1.2 Preventive Maintenance

The followings are preventive items and recommended frequency.

3.1.3 Recommended Frequency

The following tasks must be completed by professional maintenance personnel approved by our company. The equipment must be cleaned and disinfected before testing or maintenance.

Inspection/ Maintenance Items		Frequency
Complete machine's appearance inspection		1. Once every year; 2. When first installation, or after each reinstallation and maintenance.
Startup test		1. Once every year; 2. When first installation, or after reinstallation; 3. After each maintenance or main accessory replacement.
Accuracy calibration		1. Once every year; 2. When first installation; 3. When you use a new brand of infusion set; 4. After each maintenance or main accessory replacement.
Pressure calibration		1. When you use a new brand of non-built in infusion set; 2. After pressure sensor is replaced.
Operational test		1. Once every year; 2. When you doubt that the running of infusion pump and its alarm function are abnormal.
Battery powered test		1. Once every two months; 2. When first installation, or after reinstallation; 3. After maintenance of the infusion pump or when you suspects that the battery is the failure source.
Electrical safety test	Housing leakage current test	After maintenance of the power module.
	Patient leakage current test	
Cleaning and disinfection		1. Once every month. 2. Before and after long-time storage.

3.2 Complete Machine's Appearance Inspection

- 1) Clean housing, no scratch, assembled firmly and no abnormal sound when shaken.
- 2) Good handling with the keys.
- 3) Complete and correct labels.
- 4) Complete standard configuration and firm socket installation.

3.3 Startup Test

1. Press , the system will initiate the self-test and the screen will display the **[System Self-test]** interface.
2. The loudspeaker will give out a sound "di", and buzzer will give out a sound "didi".
3. The color of the alarm light will change from red to yellow, turn on and off orderly.
4. Enter the operation interface after successfully completing the system self-testing, and now you can manually operate the system through the key board.

NOTE

-
- During startup, self-test will be performed in the following sequence : 1 Self-test of CPU when powered on; 2 Loudspeaker; 3 Communication test (drives, power and keys) ; 4 Alarm light ; 5 Buzzer; 6 Up and downward pressure sensor detection.
 - If the system self-test fails, press  to enter the system.
-

3.4 Accuracy Calibration

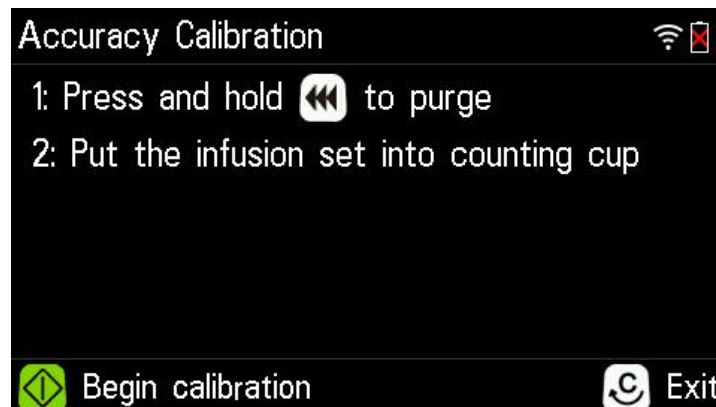
Accuracy of infusion pump requires no daily calibration. However, calibration is required when the infusion pump is used for the first time, replaced with a new brand of infusion set or when you doubt that the deviation of the injection flow is much larger.

Prepare the following materials before the calibration:

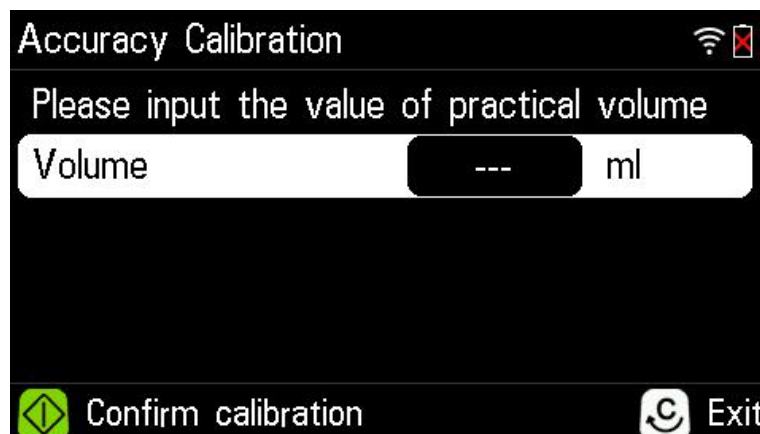
- One electronic scales (500g)
- One 100ml beaker
- One infusion set

The steps for accuracy calibration are as follows:

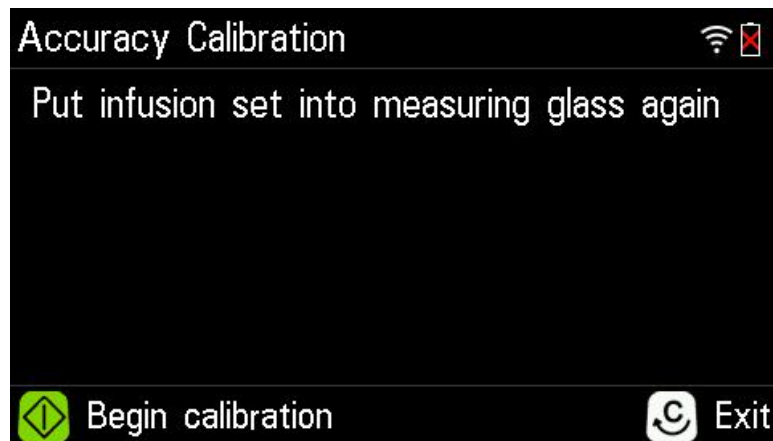
1. Connect the infusion set to the infusion bottle (bag) and install the set onto the infusion pump.
2. Enter the **[Accuracy Calibration]** interface: Select **[Main Menu]**→**[System Maintenance]**→Input Maintenance Password→**[Accuracy Calibration]**.
3. Interface will display the prompt information as shown in the figure, select the brand of infusion set as prompt.



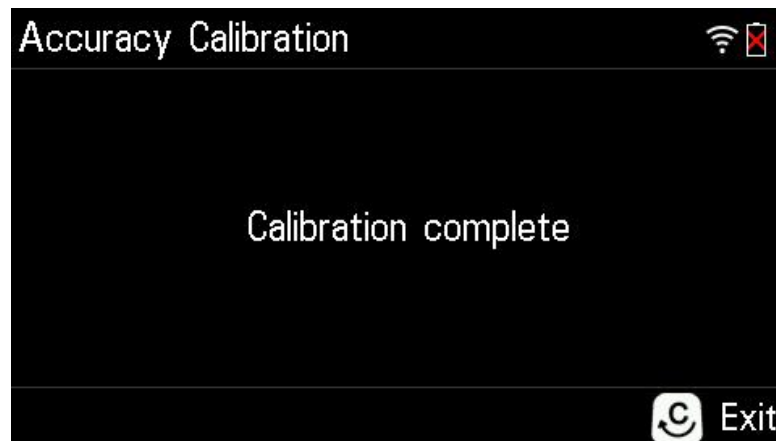
4. Press and hold [Bolus icon] to purge, put the ending of the infusion set into beaker.
5. Press [Start icon] to start the automatic calibration of the infusion pump.
6. Weigh the liquid in beaker after calibration is completed, input the actual volume to **[Volume]**.



7. Press [Start icon] to confirm data and enter to the next calibration interface, put the ending of the infusion set into beaker.



8. Press  to start the automatic calibration of the infusion pump.
9. Weigh the liquid in beaker after calibration is completed, input the actual volume to [Volume]. Press , you will be prompted with “Calibration complete”.



10. Press , calibration is completed and back to the [Main Menu] interface.

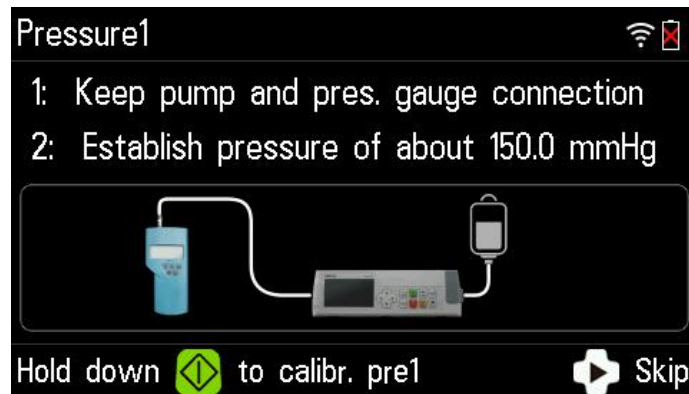
NOTE

- During accuracy calibration, please use medical normal saline solution, glucose solution or grade III water for experiments.
 - Please remove the needle portion of the infusion set, and do not place the ending of the infusion set above the liquid level in the beaker.
 - Please do not place the ending of the infusion set above the liquid level of beaker.
 - During the installation of the infusion set, please place the infusion set flat to the appropriate position and do not bend or stretch it.
 - Please do not reuse the extruded infusion set for accuracy calibration.
-

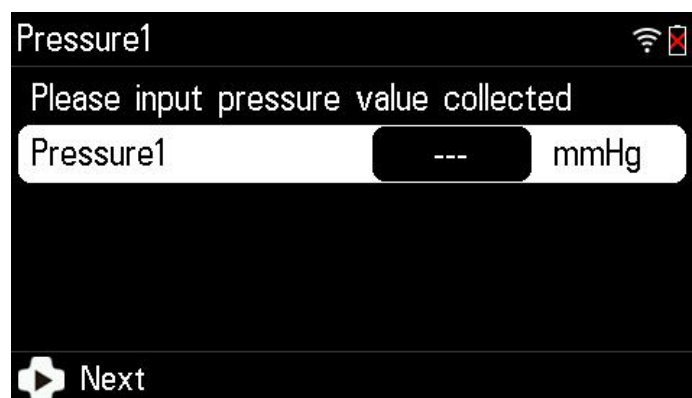
3.5 Pressure Calibration

The steps for pressure calibration are as follows:

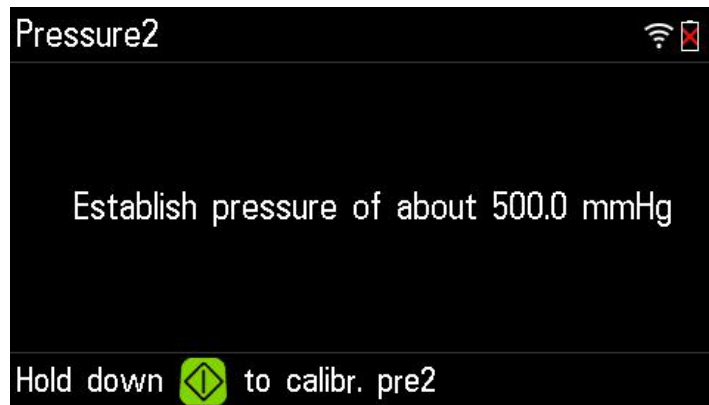
1. Enter the [Pressure Calibration] interface: Select [Main Menu]→[System Maintenance]→Input Factory Maintenance Password→[Pressure Calibration].
2. Select the brand of the infusion set as prompted.



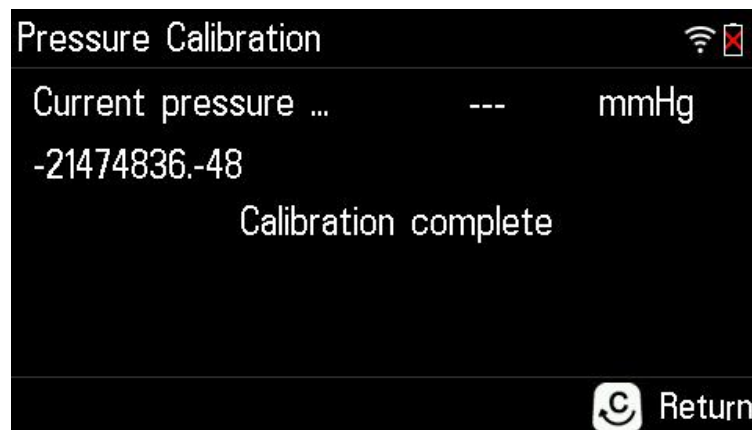
3. Connect the infusion set to the infusion bottle (bag) and install the infusion set to the infusion pump. Connect the infusion set to a pressure gauge.
4. Press  to start the first calibration. Once the pressure reading on the manometer reaches 150mmHg, press  to start collecting pressure values.
5. After the reading on the manometer stabilizes, input the pressure value as prompted, and press  for the next step.



6. Press  to start the second calibration. Once the pressure reading on the manometer reaches 500mmHg, press  to start collecting pressure values.



7. After the reading on the manometer stabilizes, input the pressure value as prompted. Press , you will be prompted with [Calibration complete].



8. Press , calibration is completed and back to the [Main Menu] interface.

WARNING

- The Pressure Calibration screen has no occlusion alarm. To avoid risk of serious harm to patients, do not perform infusion in this screen.

NOTE

- The pump cannot be switched off directly from the Pressure Calibration screen.
 - If you have doubts about the accuracy of occlusion alarm pressure (pressure value exceeding the alarm threshold), you will need to perform a pressure calibration.
 - Only pressure measuring tools recommended and approved by this Company can be used for the pressure calibration.
-

3.6 Operational Test

The steps for conventional functions test are as follows:

1. When various keys are operated on, the infusion pump shall perform or react normally in accordance with the **User's Manual**; abnormal operation should not cause the machine to crash.
2. Operate the infusion pump in accordance with the **User's Manual**. Check whether the sound-light alarm at various levels is normal, whether the volume control is normal and whether the alarm silence is working.
3. If an alarm is triggered when the pump is operated in accordance with the **User's Manual**, check whether the operations and alarms are properly recorded in the History Record.
4. Install the infusion set, set the infusion parameters correctly, and then press  to enter the running interface.
5. During infusion, the operation icon, rate, volume and pressure value shall be normal.

The steps for networking function test are as follows (Only for infusion pumps with wireless network function):

1. Regardless of which one of BeneFusion CS5 Infusion Supervision System or the pump is powered on first, networking shall always work.
2. Transmission of the infusion data to BeneFusion CS5 Infusion Supervision System is normal.
3. Information on the pump status, infusion parameters and alarm received by BeneFusion CS5 Infusion Supervision System is correct.
4. After the network is disconnected at a certain link, BeneFusion CS5 Infusion Supervision System can correctly indicate the disconnection; when the link is reconnected, networking and work are back to normal.

3.7 Battery Power Test

The steps for battery power test are as follows:

1. System can give a prompt when no battery is installed (See **User's Manual** for detailed alarm information).
2. After "Battery Empty" alarm is triggered, a pump in operation or a paused pump during operation may not be started.

Note: See **User's Manual** for detailed information of battery maintenance.

3.8 Electrical Safety Test

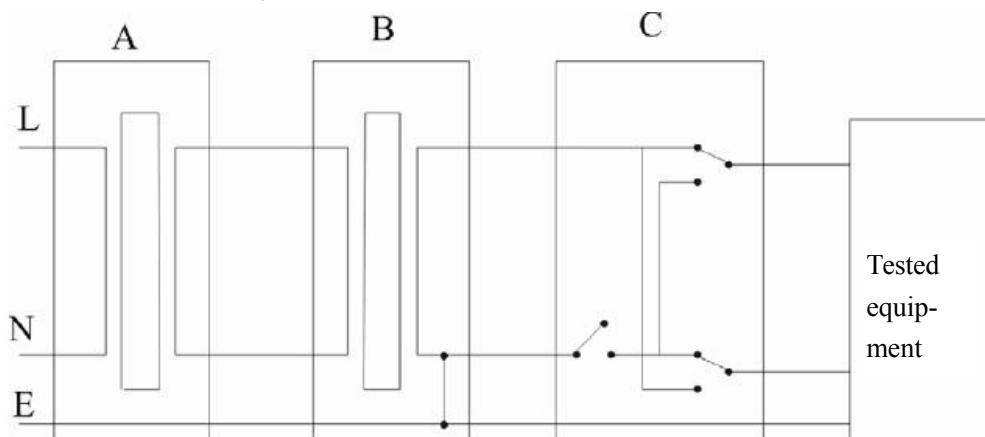
WARNING

- To test the electrical safety of the infusion pump, the electrical safety test is designed to detect abnormal electrical hazards which, if not discovered, may cause injury to the patient or the operator.
 - Commercially available testing devices such as the safety analyzer may be used for electrical safety tests. The maintenance personnel shall ensure the applicability, functional integrity and safety of such devices and familiarize themselves with the use of these devices.
 - Electrical safety test shall follow the following standard: IEC/EN 60601-1.
 - If local regulations provide provisions otherwise, please perform the relevant electrical safety test in accordance with the provisions.
 - In the patient area, all devices that are connected to the mains supply as well as to medical equipment must comply with the IEC/EN 60601-1 standard, and must be tested for electrical safety in accordance with the test interval for monitoring devices.
-

Electrical safety test is used to detect hazards that may pose electrical safety threats to the patient, operator and maintenance personnel. Please perform the electrical safety test in normal environment (including temperature, humidity and atmospheric pressure).

While the 601 safety analyzer is used as an example in the electrical safety test described in this chapter, different safety analyzers may be used in different regions. Please ensure the applicability of the electrical safety test you are going to perform.

Device connection diagram is as follows:



A: AC power supply (programmable power supply, regulating frequency)

B: Isolation transformer on the leakage current test tool

C: Safety tester

Testing tools:

■ Safety analyzer

■ Isolation transformer

3.8.1 Housing Leakage Current Test

1. Connect the 601 safety analyzer to a power supply of 264 VAC and 60 Hz.
2. Using the connection tooling of the application section, connect the application section of the tested device and connect the SUM end of the connection tooling of the application section to the RA end of the safety analyzer.
3. Connect the tested device, via the power line, to the auxiliary power output jack of the 601 safety analyzer.
4. Connect one end of the red test lead to the "Red input terminal" of the safety analyzer and clip the other end to the metal foil attached to the surface of the housing of the tested device.
5. Power on the 601 safety analyzer, and press "5-Enclosureleakage" on the panel of the 601 safety analyzer to enter the interface for the housing leakage current test.
6. Housing leakage current is no more than 100 μ A under normal condition and no more than 300 μ A under single-fault condition.

3.8.2 Ground Leakage Current Test

1. Connect the 601 safety analyzer to a power supply of 264 VAC and 60 Hz.
2. Connect the application section of the tested device to the RA end of the safety analyzer.
3. Connect the tested device, via the power line, to the auxiliary power output jack of the 601 safety analyzer.
4. Power on the 601 safety analyzer, and press "4-Earth leakage" on the panel of the 601 safety analyzer to enter the interface for the ground leakage current test.
5. Housing leakage current is no more than 500 μ A under normal condition and no more than 1000 μ A under single-fault condition.

3.8.3 Patient Leakage Current Test

1. Connect the 601 safety analyzer to a power supply of 264 VAC and 60 Hz.
2. Using the connection tooling of the application section, connect the application section of the tested device and connect the SUM end of the connection tooling of the application section to the RA end of the safety analyzer.
3. Connect the tested device, via the power line, to the auxiliary power output jack of the 601 safety analyzer.
4. Power on the 601 safety analyzer, and press "6-Patient leakage" on the panel of the 601 safety analyzer.
5. Continuously press the "APPLIED PART" key to select the AC and DC measurements; "DC" is shown following the limit value of direct current.
6. Patient leakage current is no more than 10 μ A under normal condition and no more than 50 μ A under single-fault condition.

If the electrical safety test fails, please contact our after-sale service department.

3.9 Cleaning and Disinfection

3.9.1 Cleaning

The pump should be cleaned regularly. If operating in dirty or sandy areas, cleaning should be more frequent. Before cleaning, please consult or refer to the hospital's specific regulations concerning medical device cleaning.

The following are detergents available for selection:

- Hydrogen peroxide (3%)
- Ethanol (70%)
- Isopropanol (70%)

When cleaning the device:

1. Turn off the pump and disconnect the AC power source line.
2. Wipe the display screen after soft cotton balls absorb an appropriate amount of detergent.
3. Use a piece of soft cloth which absorbs a modest amount of cleaning agent to wipe the surface of the device.
4. When necessary, use a piece of cloth to wipe off any excess cleaning agents.
5. Place the equipment in a cool and ventilated environment to dry.

3.9.2 Disinfection

The operation of disinfection may cause certain damage to the infusion pump. You are recommended to disinfect only when it is necessary in your desired maintenance plan. Clean the equipment before disinfection.

The recommended disinfectants include: ethano 70%, isopropanol 70%, glutaraldehyde-type 2% liquid disinfectant.

CAUTION

- **Never use EtO or formaldehyde for disinfection.**
 - **Do not conduct high pressure or high temperature disinfection for the infusion pump and its accessories.**
-

4 Troubleshooting

4.1 Overview

In this chapter, faults of infusion pump are classified according to the faulty components and faulty phenomena. Please refer to the relevant Fault Table when fault examination, identification and troubleshooting in sequence.

The recommended solutions given in this chapter should help you solve most of the equipment faults you will encounter but not all possible problems. In the case of a fault not covered in this chapter, please contact our After-sale Service Department.

4.2 Replacement of Components

You may replace the circuit board components and other major components or parts of this infusion pump. Once you have identified the faulty circuit board component, you may follow the steps described in **Chapter 5 Maintenance and Disassembly** to replace the circuit board component. Then you may check whether the fault has been eliminated or whether the infusion pump is able to pass the relevant tests. If the fault has been eliminated, which shows that the original circuit board is damaged, and then please return the faulty circuit board component to our company for repair. If the fault persists, please reassemble the original circuit board for troubleshooting according to other possible causes.

Please refer to **Chapter 6 Components** for more information about the components to be replaced.

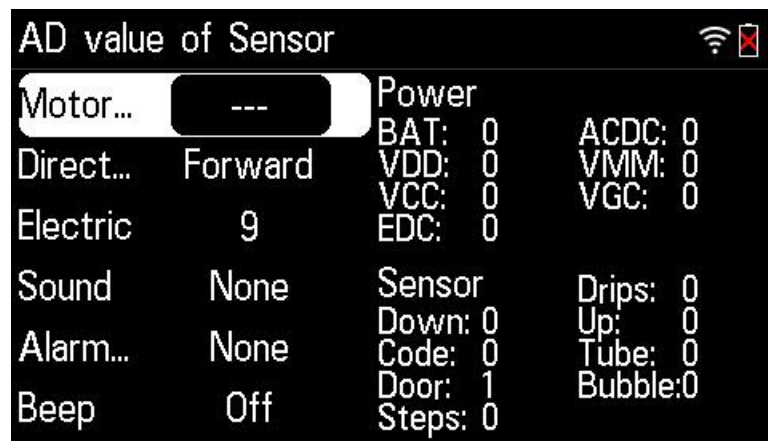
4.3 Check the Infusion Pump Status

4.3.1 Operation Methods for AD Value Channels and the

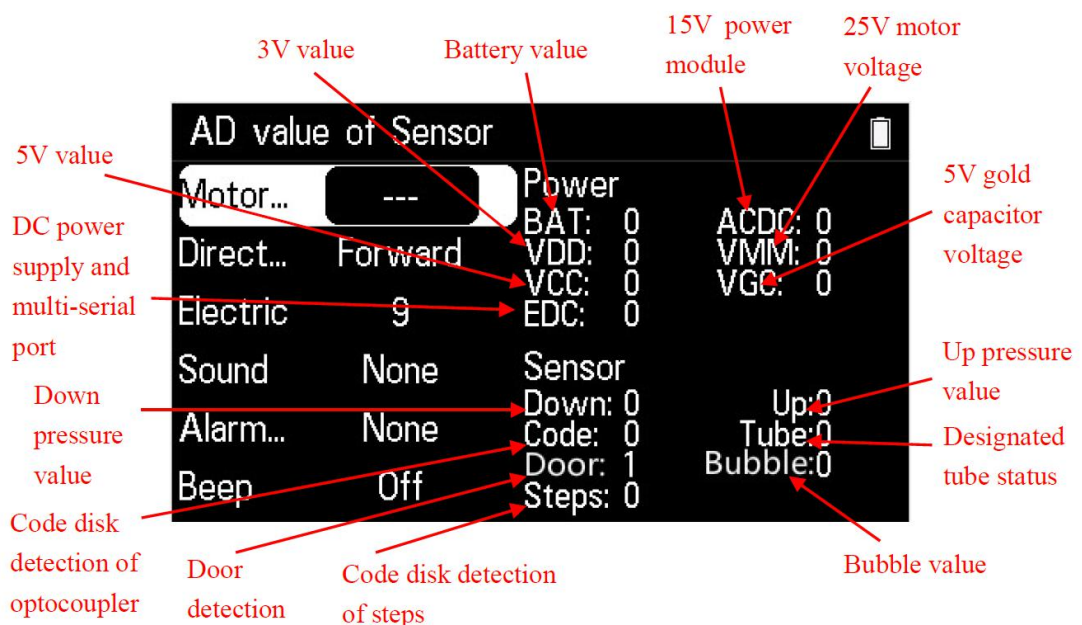
Corresponding Tests

1. Operation method for entering the AD value interface: System maintenance --- enter the password "6686" --- sensor AD value.

2. AD value interface is as follows:



3. The meaning of each AD value represents is demonstrated in the diagram below:



4. Down indicates the output AD value by the downward pressure sensor; when the downward pressure sensor exerts pressure, this value should increase.

5. Up indicates the output AD value by the upward pressure sensor; when the upward pressure sensor exerts pressure, this value should increase.

6. Code indicates the output AD value by the code disk detection of optocoupler; under normal circumstances, this value should be changed among 1, 2 and 3 when the motor is running.

7. Door's value indicates the AD values of the door detection sensor and liquid check clip detection sensor.

-
- When the infusion pump door is open and the liquid check clip is open, Door's value is 3.
 - When the infusion pump door is open and the liquid check clip is closed, Door's value is 1.
 - When the infusion pump door is closed and the liquid check clip is closed, Door's value is 0.

8. Tube indicates designated infusion set on-line detection (only applicable to closed models of consumables).

9. Bubble indicates the bubble sensor value which increases after the infusion set is installed. Under normal circumstances, the value is greater than 2bc when the infusion set is filled with water and less than 96 when the infusion set is empty.

10. Steps indicates the code disk detection of steps, the more the operational steps, the greater the value.

4.4 Fault Table

During transportation, storage and use of infusion pump, there can be faults that may affect the normal use as a result of various factors (e.g. grid voltage instability, change in ambient temperature, drop and shock and component aging). In such cases, professional personnel with certain experience in electronic medical equipment maintenance shall carry out component level overhauling in accordance with the following fault classification. The so-called part level overhauling refers to the analysis, replacement, test-run and identification of the equipment's faulty component(s), such as power supply, main control section, monitoring section and/or LCD. The maintenance of components, also known as component level maintenance, must be carried out by the manufacturer's engineers with rich experience in maintenance, using specialized equipment, in a specific environment and under specific conditions.

4.4.1 Alarm Information and Countermeasures

Please refer to ***BeneFusion VP5 User's Manual*** for details.

4.4.2 Complete Machine Faults

Faulty phenomenon	Possible cause	Handling method
When external power supply is not connected, there is no display on startup, battery indicator is not lit and buzzer does not sound.	1. Battery is dead; 2. Bad connection of the battery interface with the wires; 3. Bad connection of the keyboard with the FPC wires; 4. Faulty main control board.	1. Battery replacement; 2. Repair or replacement of the battery interface with the wires; 3. Re-installment or replacement of FPC wires; 4. Replacement of the main control board.
When external power supply is not connected, there is no display on startup, battery indicator is lit and buzzer sounds.	1. Faulty Display; 2. Short circuit.	1. Treatment as with Faulty Display; 2. Further examination of the main control board and other boards.
When external AC power supply is connected, AC indicator is not lit.	1. Bad connection of the keyboard with the FPC wires; 2. Bad connection of the AC power supply wires; 3. Faulty ACDC module.	1. Repair or replacement of FPC wires; 2. Repair or replacement of AC power supply wires; 3. Repair or replacement of ACDC module;
Self-test 3 not passed; alarm is triggered.	1. Bad connection of the keyboard with the FPC wires; 2. Faulty main control board; 3. Faulty Keyboard.	1. Repair or replacement of FPC wires; 2. Replacement of the main control board; 3. Replacement of the Keyboard.
Self-test 5 not passed; alarm is triggered.	1. Faulty power supply of the main control board; 2. Short circuit of the board.	1. Replacement of the main control board; 2. Check each board whether there is short circuit.
Self-test 6 not passed; alarm is triggered.	1. Faulty upward pressure detection; 2. Faulty downward pressure detection.	1. Repair upward pressure sensor wires, or replacement of upward pressure sensor or main control board; 2. Repair downward pressure sensor wires, or replacement of downward pressure sensor or detection plate, or repair downward pressure detection plate to main control board wires.

Unable to charge properly	1. Bad connection of the battery interface with the wires; 2. Faulty battery; 3. Faulty main control board.	1. Repair or replacement of the battery interface with the wires; 2. Replacement of the battery; 3. Replacement of the main control board.
Failure of an operation or measurement function	Damage to the main control board or its parts.	Further examination of the main control board and its parts.

4.4.3 Display Faults

Faulty phenomenon	Possible cause	Handling method
Power supply normal on startup, black or white Display Screen; Black or white Display Screen during normal operation.	1. Bad connection of the keyboard with the FPC wires; 2. Faulty Display Screen; 3. Faulty Keyboard; 4. Faulty main control board.	1. Repair or replacement of FPC wires; 2. Replacement of the Display Screen; 3. Replacement of Keyboard; 4. Replacement of the main control board.
Error in displayed texts or images.	1. Error in burning resource files; 2. Faulty main control board.	1. Burn resource files again; 2. Replacement of the main control board.

4.4.4 Operation and Alarm Faults

Faulty phenomenon	Possible cause	Handling method
Key failure	1. Bad connection of the keyboard with the FPC wires; 2. Faulty Keyboard.	1. Repair or replacement of FPC wires; 2. Replacement of Keyboard.
Buzzer sound failure	Faulty main control board.	Replacement of the main control board.
Speaker sound failure or hoarse sound	1. Faulty speaker; 2. Bad connection of the battery interface with the wires; 3. Faulty alarm circuit of the speaker.	1. Repair of the wires or replacement of the speaker's components; 2. Repair or replacement of the battery interface with the wires;

		3. Further examination of the faulty circuit and replacement of the main control board.
Failure of the red alarm light or yellow alarm light	1. Faulty alarm light plate; 2. Bad connection of the keyboard with the FPC wires; 3. Faulty Keyboard; 4. Faulty main control board.	1. Repair of the wires or replacement of the alarm light plate; 2. Repair or replacement of FPC wires; 3. Replacement of the Keyboard; 4. Replacement of the main control board.
Failure of the key backlight	1. Bad connection of the keyboard with the FPC wires; 2. Faulty Keyboard; 3. Faulty main control board.	1. Repair or replacement of FPC wires; 2. Replacement of the Keyboard; 3. Replacement of the main control board.

4.4.5 Monitoring Faults

Faulty phenomenon	Possible cause	Handling method
Pressure value unchanged	1. Failure of the downward pressure sensor; 2. Failure of the downward pressure detection plate; 3. Loose or damaged wires of downward pressure detection plate and main control board.	1. Replacement of the downward pressure sensor; 2. Replacement of the downward pressure detection plate; 3. Repair or replacement of wires of downward pressure detection plate and main control board.
Pressure value overrange	1. Loose or damaged wires of downward pressure detection plate and main control board. 2. Failure of the downward pressure sensor; 3. Failure of the downward pressure detection plate.	1. Repair or loose wires of downward pressure sensor; 2. Replacement of the downward pressure sensor; 3. Replacement of the downward pressure detection plate.

Upstream occlusion alarm is triggered mistakenly	1. Failure of the upward pressure sensor; 2. Failure of the upward pressure detection plate.	1. Replacement of the upward pressure sensor; 2. Replacement of the main control board.
Door opened alarm is triggered mistakenly	Loose or damaged door detection limit switch.	Repair or replacement of the limit switch.
Air in line alarm is triggered mistakenly	1. Loose wires of the ultrasonic transmitter chip or damaged ultrasonic transmitter circuit; 2. Loose wires of the ultrasonic receiver chip or failure of the ultrasonic detection circuit.	1. Repair of the wires of the ultrasonic transmitter chip or replacement of the keyboard; 2. Repair of the wires of the ultrasonic receiver chip or replacement of the bubble detection plate.
Motor Error	1. Failure of the motor detection plate; 2. Bad connection of the motor wires; 3. Faulty drive circuit for the stepper motor; 4. Motor error.	1. Replacement of the motor detection plate; 2. Repair or replacement of the motor wires; 3. Replacement of the main control board; 4. Replacement of the motor.
Faulty liquid check clip	1. Faulty liquid check clip detection circuit; 2. Loose or damaged wires of downward pressure detection plate and main control board.	1. Replacement of the downward pressure detection plate; 2. Repair or replacement of wires of downward pressure detection plate and main control board.
Faulty designated infusion set	1. Faulty detection circuit of designated infusion set; 2. Loose or damaged wires of downward pressure detection plate and main control board.	1. Replacement of the downward pressure detection plate; 2. Repair or replacement of wires of downward pressure detection plate and main control board.

5 Maintenance and Disassembly

NOTE

- During maintenance and disassembly, an anti-static wrist strap should be worn at all times.
-

5.1 Tools

During disassembly and replacement of components, you may need the following tools:

- ◆ Phillips screwdriver
- ◆ Tweezers
- ◆ Needle-nose pliers
- ◆ Diagonal pliers
- ◆ Anti-static wrist strap

5.2 Preparation for Disassembly

Before disassembly of the infusion pump, please stop infusion to the patient, turn off the infusion pump, and disconnect all accessories and external equipment.

CAUTION

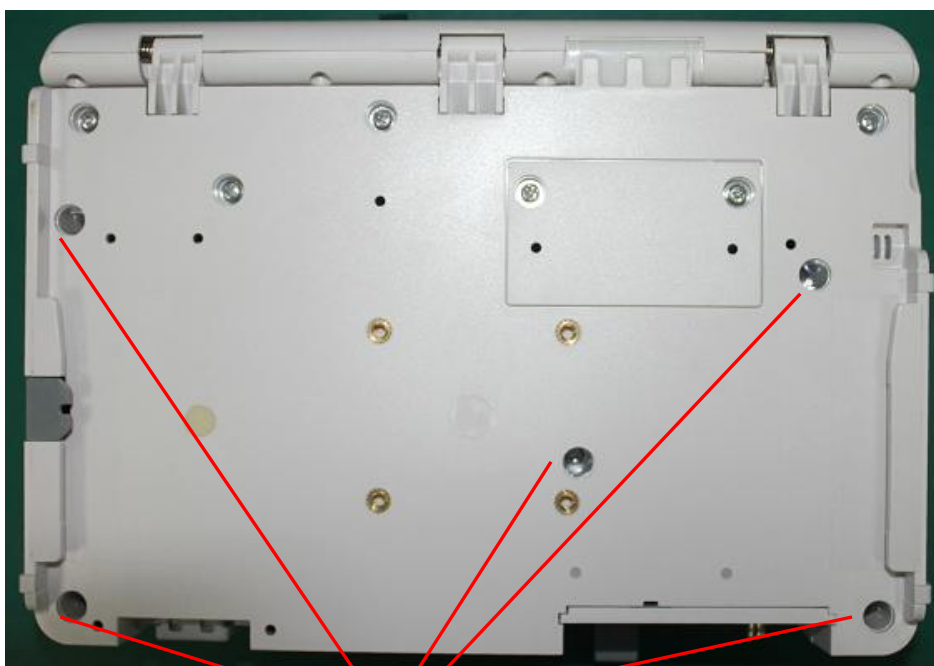
- Maintenance personnel should eliminate static prior to disassembly. During disassembly of the parts of certain static-sensitive devices, protective gears such as anti-static wrist strap or gloves should be worn to avoid damage to the parts.
 - During re-installation, please make sure the connection wires are plugged in and properly placed to avoid the wires being crushed which may cause a short circuit.
 - Note that suitable screws should be used during re-installation. If incorrect screws are screwed in by force, equipment damage may ensue; during the operation after re-installation, screws or parts may fall off and cause unpredictable damage to the product or personal injury.
 - A certain order must be followed during disassembly. Instead if the equipment is disassembled not in this order and is done forcibly, irreversible damage to the equipment may ensue.
-

-
- When components are disassembled, please ensure that all connection wires are unplugged and avoid breakage or damage to the connectors.
 - Please place the screws and other parts and components in separate places in order to find and use them during re-installation, meanwhile avoid falling, contamination or loss.
 - During disassembly, please retain and protect the waterproof seals which should not be forgotten or damaged during installation.
-

5.3 Disassembly Procedures

5.3.1 Disassembly of the Upper Housing Assembly

1. Reverse the machine to the direction as indicated in the diagram. Unscrew and remove the 5 M3x8 Phillips pan head screws connecting the upper housing assembly to the housing base.

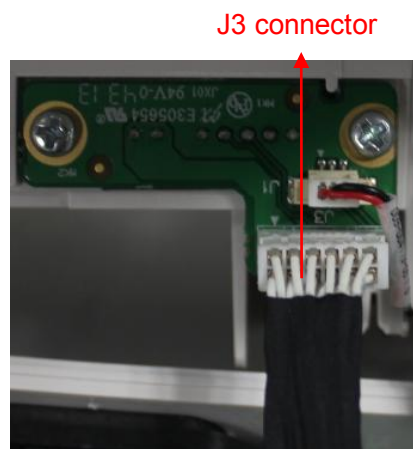


Phillips pan head screws and M3x8 dual pads

2. Vertically and upwardly pull open the housing assembly as indicated in the following diagram.



3. Unplug the wires of the battery adapter plate from the J3 connector on the battery adapter plate PCBA as indicated in the following diagram.

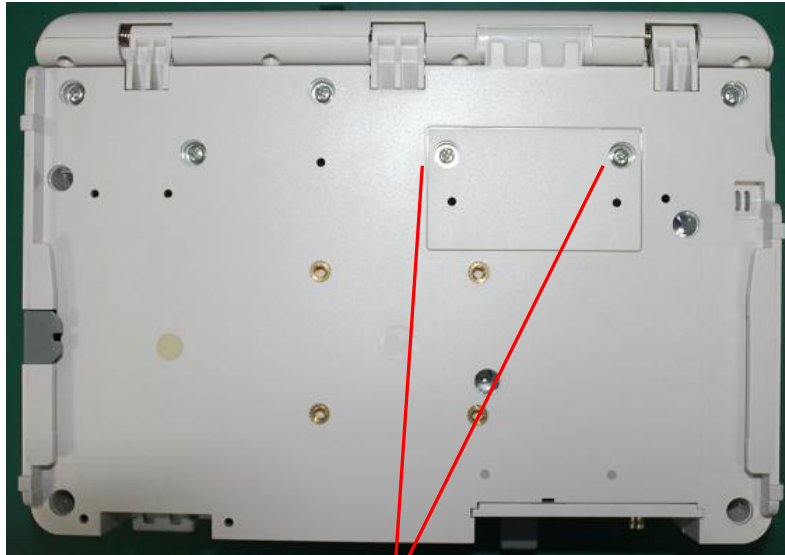


4. Now, the upper housing assembly can be removed from the host and disassembly of the upper housing assembly is completed.

5.3.2 Disassembly of the Pump Assembly and the Housing

Base Assembly

1. Remove the 2 M3x8 dual padded screws from the patch wire cover at the bottom of the housing.



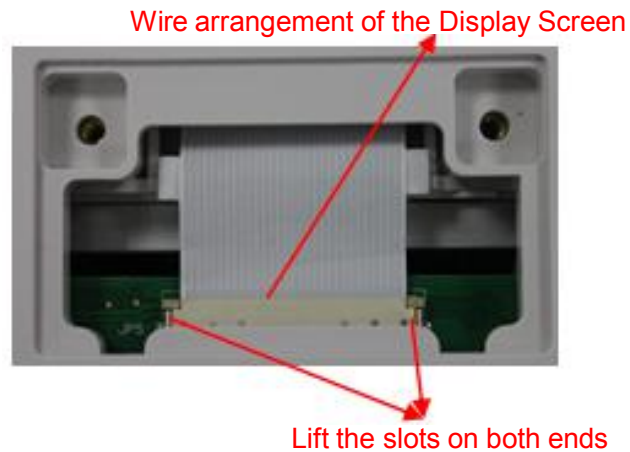
Phillips pan head screws and M3x8 dual pads

2. Remove the patch wire cover as indicated in the diagram.



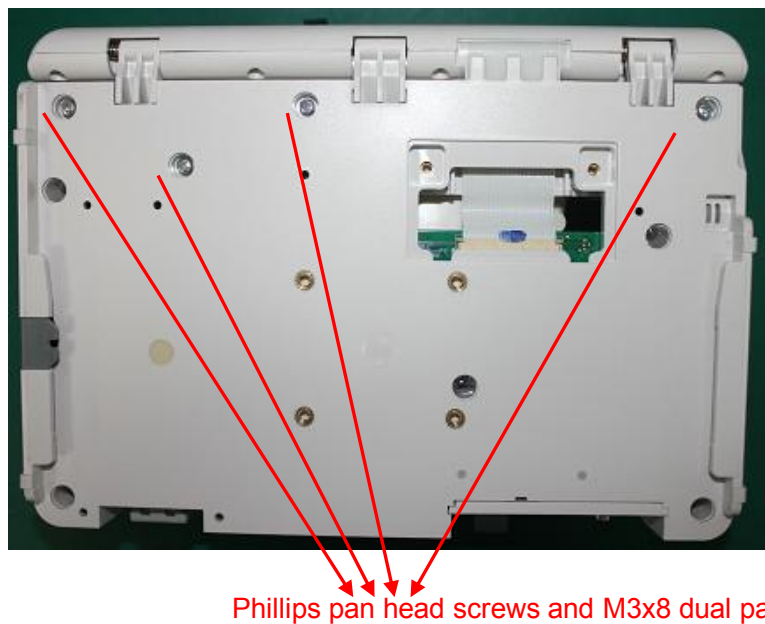
There are snaps at both ends

3. Lift the socket slots on both ends of the connector as indicated in the diagram and pull out the wire arrangement along the direction of the connector.



4. Pull out the downward pressure sensor wire, travel switch wire, code disk detection wire, motor connection wire, bubble detection connection plate and pressure detection plate connection wire from main control board.

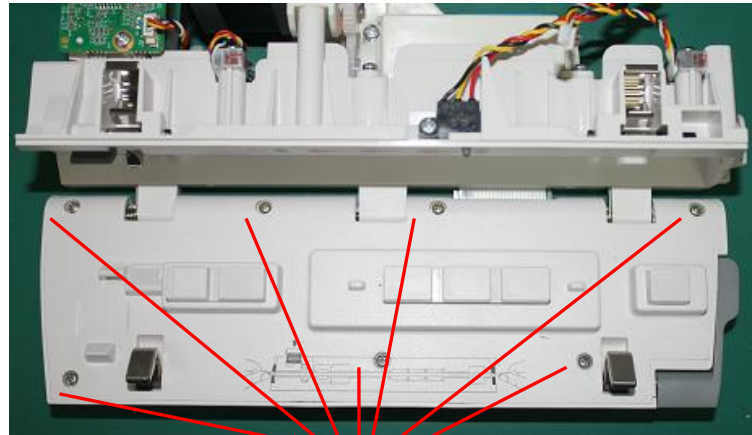
5. Reverse the machine to the direction as indicated in the diagram. Unscrew and remove the 5 M3x8 Phillips pan head screws fixing the pump assembly to the housing base.



6. Now, the pump assembly can be separated from the housing base. Disassembly of the pump assembly is completed.

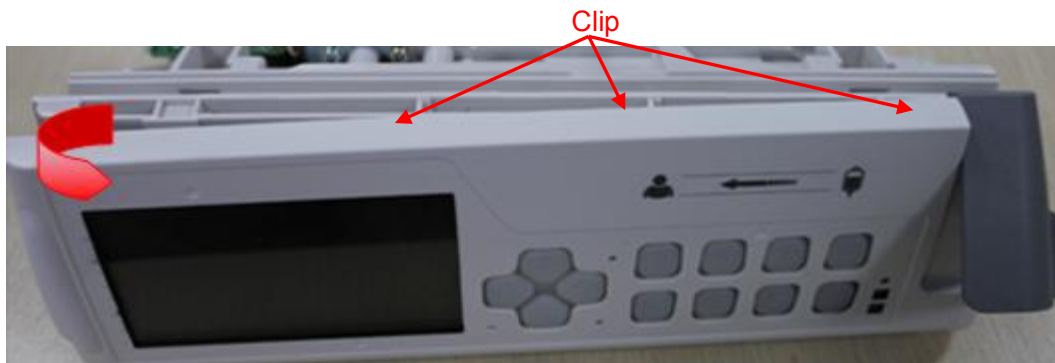
5.3.3 Disassembly of the Front/Rear Door Housing Assembly and Pump Transmission Assembly

1. After opening the infusion pump door, remove the 7 PT2.6×6 stainless steel self-tapping screws inside of the door.

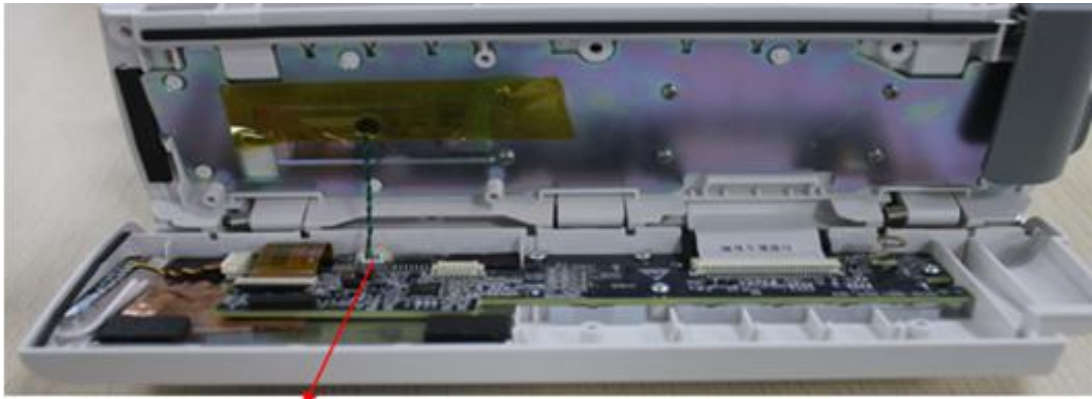


PT2.6×6 stainless steel self-tapping screws, under which a transparent nylon pad (2.5×5×0.5) is placed

2. Separate the front door housing from the rear door housing as indicated in the following diagram.

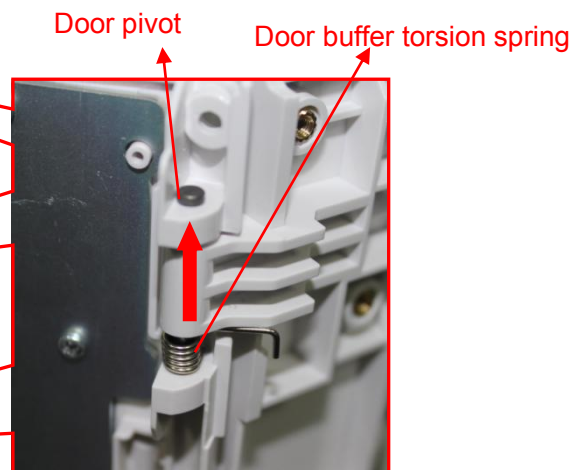
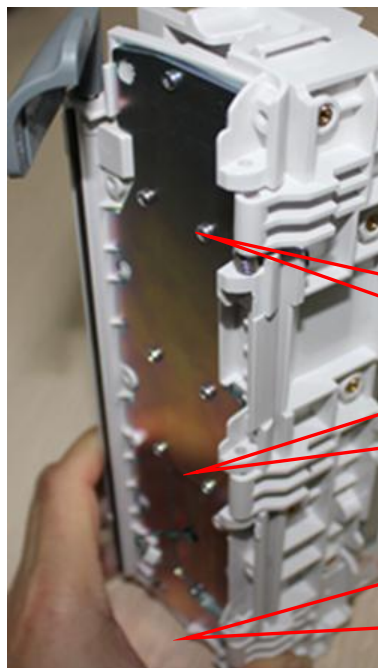


3. Unplug the ultrasonic wires from the keyboard and the front door housing assembly is separated from the pump assembly. Disassembly of the front door housing assembly is completed.



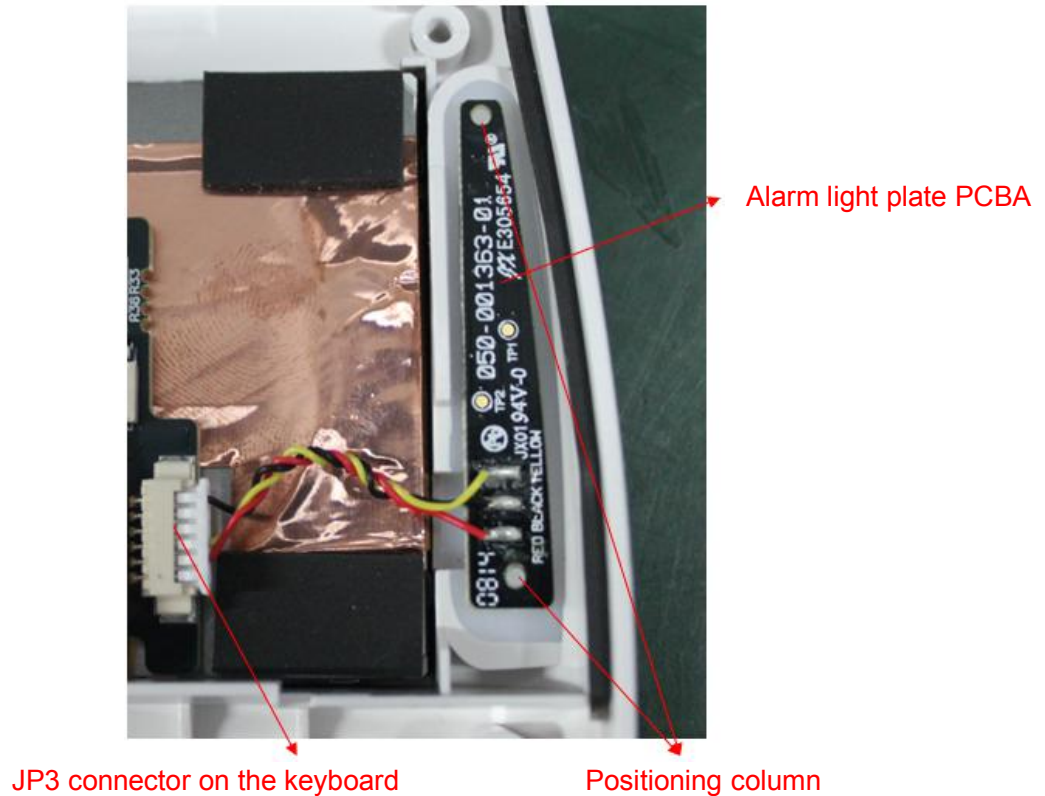
Unplug the ultrasonic wires from the J1 socket on the keyboard and plug

4. Remove the door pivot connecting the rear door housing assembly and pump transmission assembly in the direction as indicated by the arrow in the diagram. Pump transmission assembly is separated from the rear door housing assembly. Remove the door buffer torsion spring and the disassembly of the rear door housing assembly and pump transmission assembly is completed.

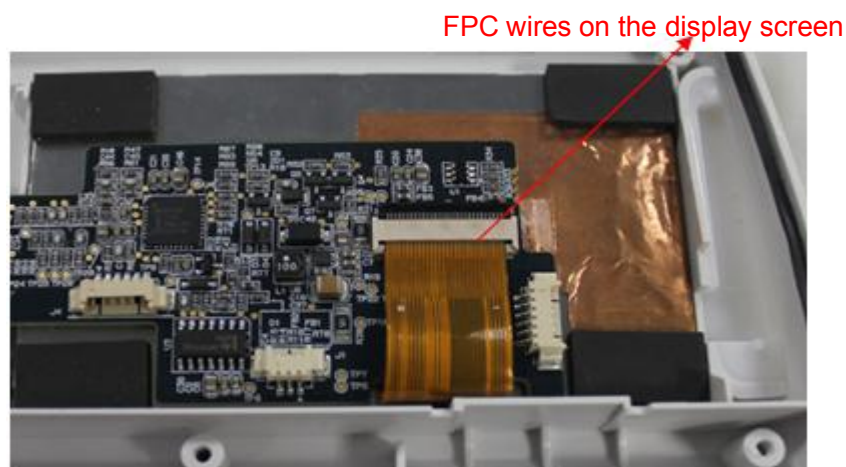


5.3.4 Disassembly of the Alarm Light Plate, Keyboard and Display Screen

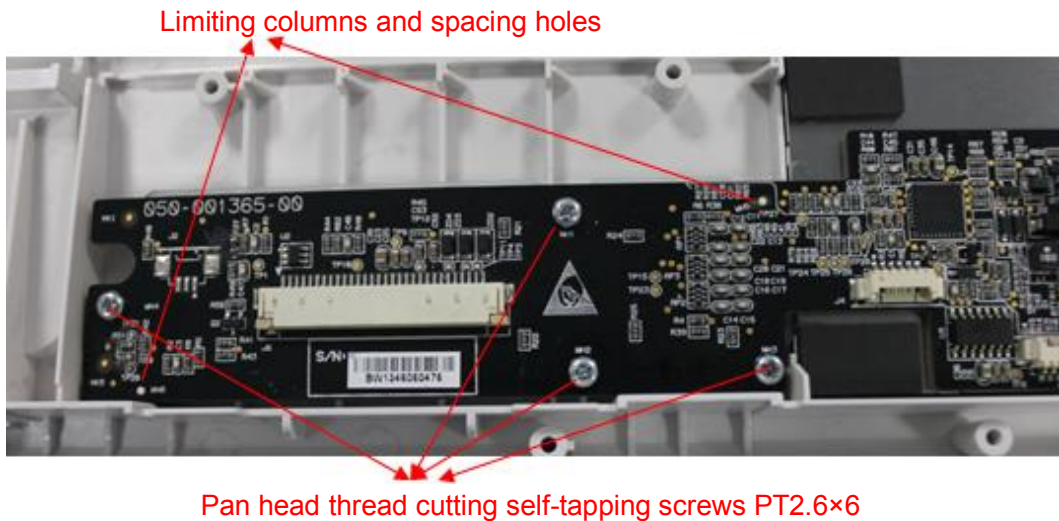
1. Unplug the alarm light wires from the keyboard and remove the alarm light plate along the direction of the positioning column. Disassembly of the alarm light plate is completed.



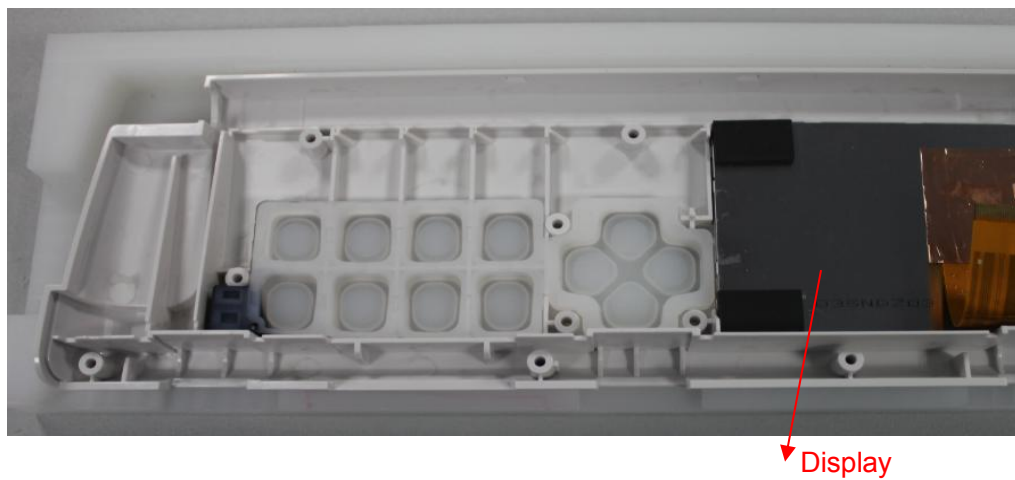
2. Unplug the FPC wires on the display screen from the keyboard.



3. Remove the 4 PT2.6×6 self-tapping screws by which the keyboard is fixed and remove the keyboard along the direction of the limiting column. Disassembly of the alarm light plate is completed.

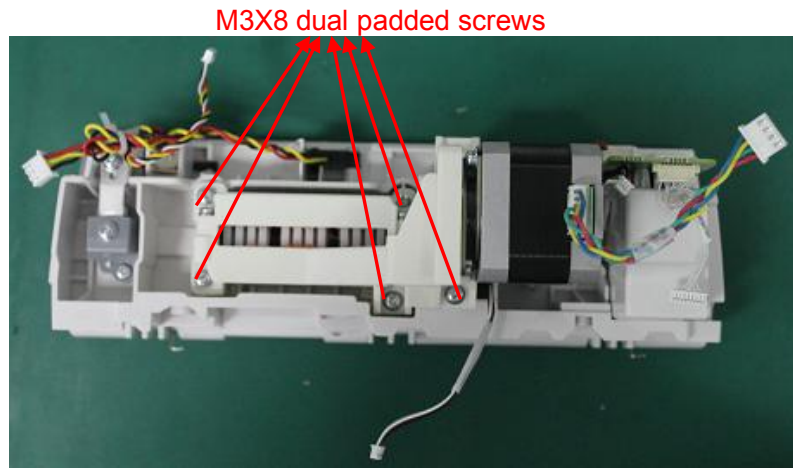


4. As indicated in the diagram, directly remove the display screen after the disassembly of the alarm light plate is completed. Disassembly of the display screen is completed.

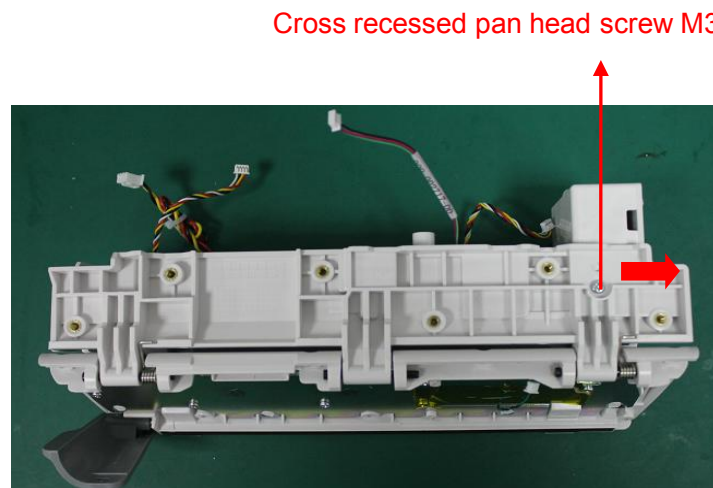


5.3.5 Disassembly of the Pressure Sensor Assembly, Pump Tablet Installation Assembly, Liquid Check Clip Assembly and Pump Ultrasonic Chip Assembly

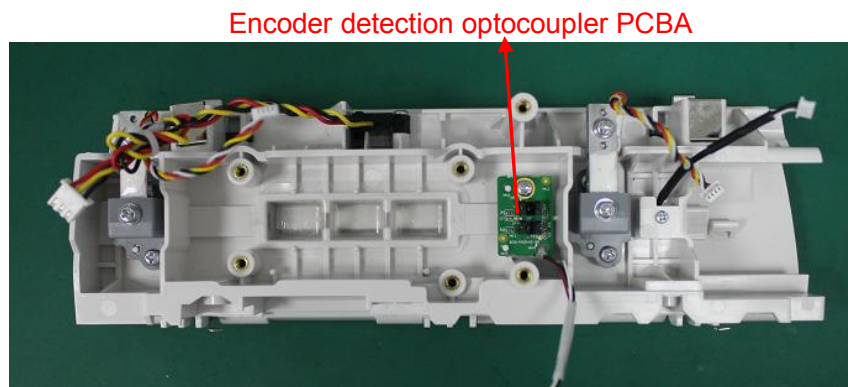
1. Remove the 5 M3X8 dual padded screws by which pump installation assembly is fixed as indicated in the following diagram.



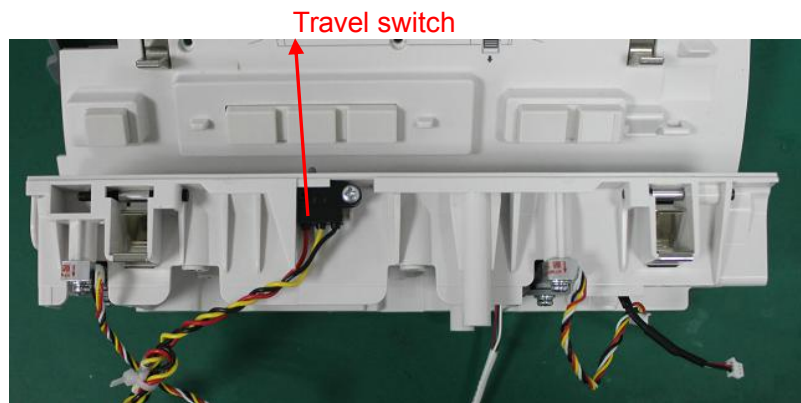
2. Pull out the connection wires of downward pressure sensor on downward pressure detection plate, remove the cross recessed pan head screw M3x14 on the base of pump by which liquid check clip assembly is fixed, and remove liquid check clip assembly in the direction of arrow. Disassembly of the liquid check clip assembly is completed.



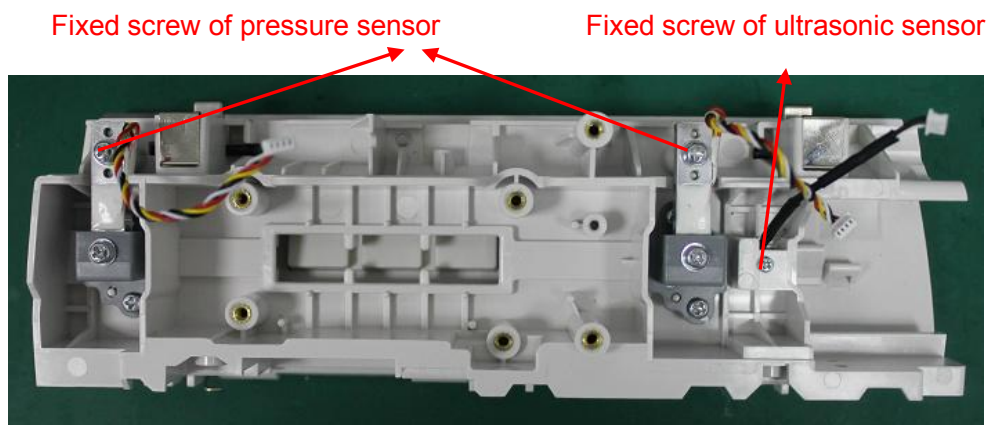
-
3. Loosen the screw on encoder detection optocoupler, and move encoder detection optocoupler PCBA.



4. Loosen the screw on travel switch, and disassemble the travel switch.

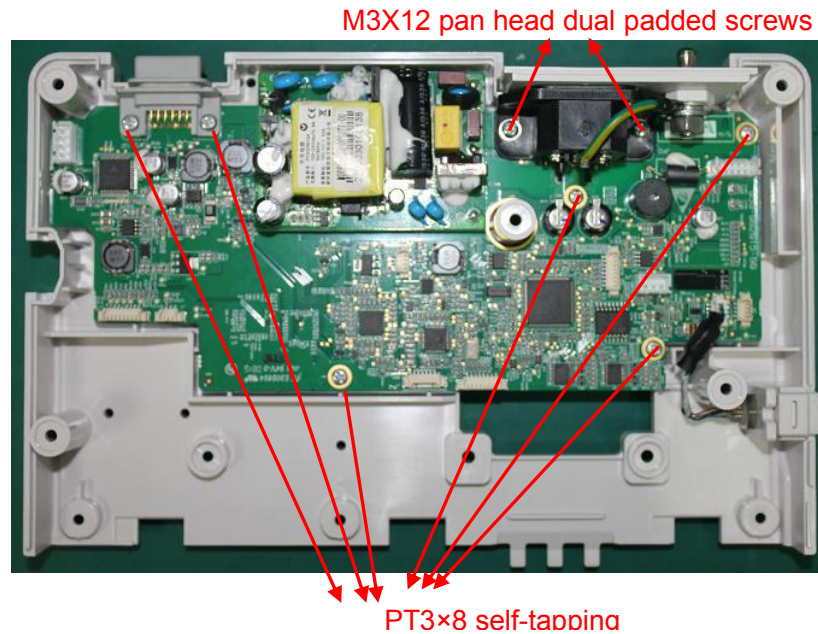


5. Remove the fixed screw of pressure sensor on the pump installation base, and disassemble the pressure sensor. Remove the fixed screw of ultrasonic sensor on the pump installation base, and disassemble the ultrasonic sensor.

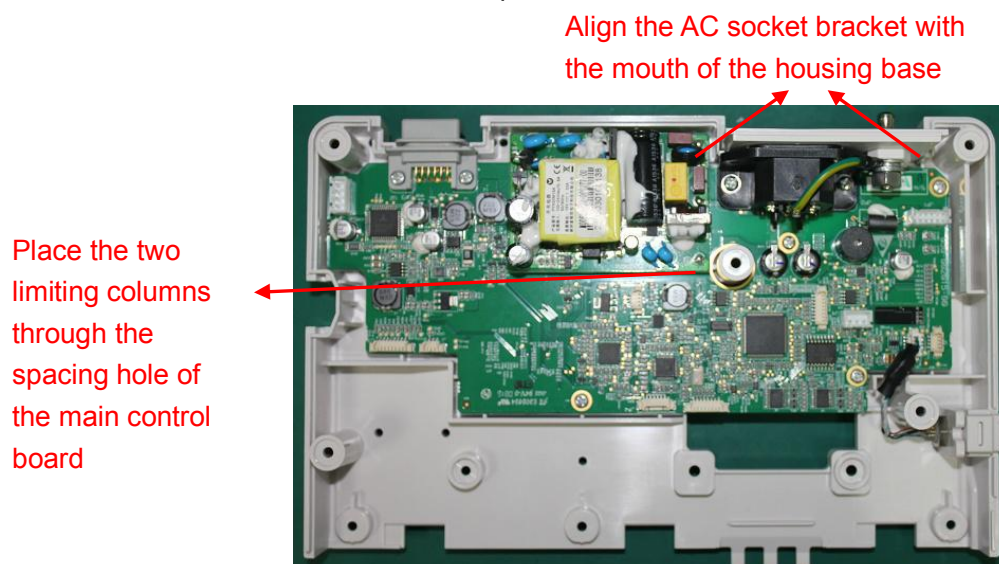


5.3.6 Disassembly of the Main Board

1. Unplug the wires on the main board PCBA.
2. Remove the 5 PT3×8 self-tapping screws and 2 M3X12 pan head dual padded screws by which the main board is fixed.



3. Remove the AC socket bracket and main control board respectively along the direction of the mouth of the housing base and the limiting columns. Disassembly of the main control board is completed.



6 Components

6.1 Description

This chapter provides the explosive view of the main components of the infusion pump host and their material codes to help the maintenance personnel identify the components during disassembly and replacement of components.

Structural system of the infusion pump host is indicated in the following figure:

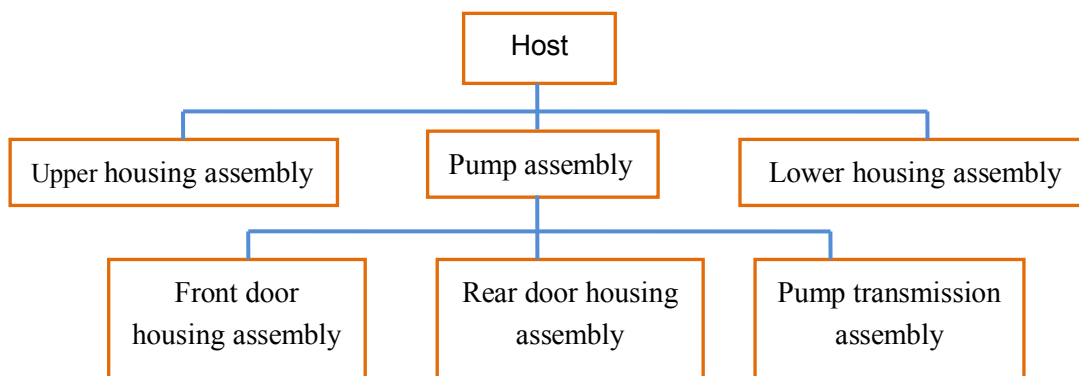


Figure 1 Structural Composition

6.2 Host

6.2.1 Explosive View

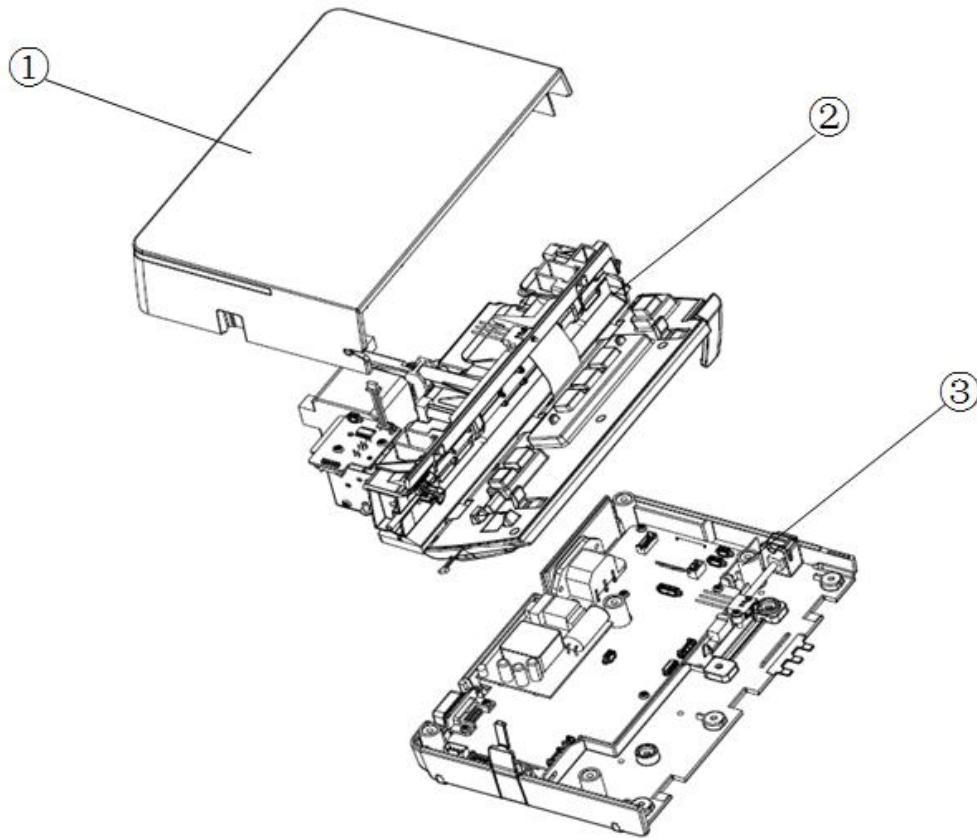


Figure 2 Explosive View of Host

6.2.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	115-038479-00	VP5 upper housing components	1
2	115-038477-00	VP5 pump body components	1
2	115-038478-00	Pump body components of VP5 ALL infusion pump with specified infusion sets	1
3	115-038480-00	VP5 bottom housing components	1
3	115-038481-00	VP5 ALL bottom housing components (with wireless)	1
3	115-038482-00	VP5 ALL bottom housing components (with drop sensor)	1
3	115-038483-00	VP5 ALL bottom housing components (with wireless & drop sensor)	1

Table 1 Parts List of Host

6.3 Upper Housing Assembly

6.3.1 Explosive View

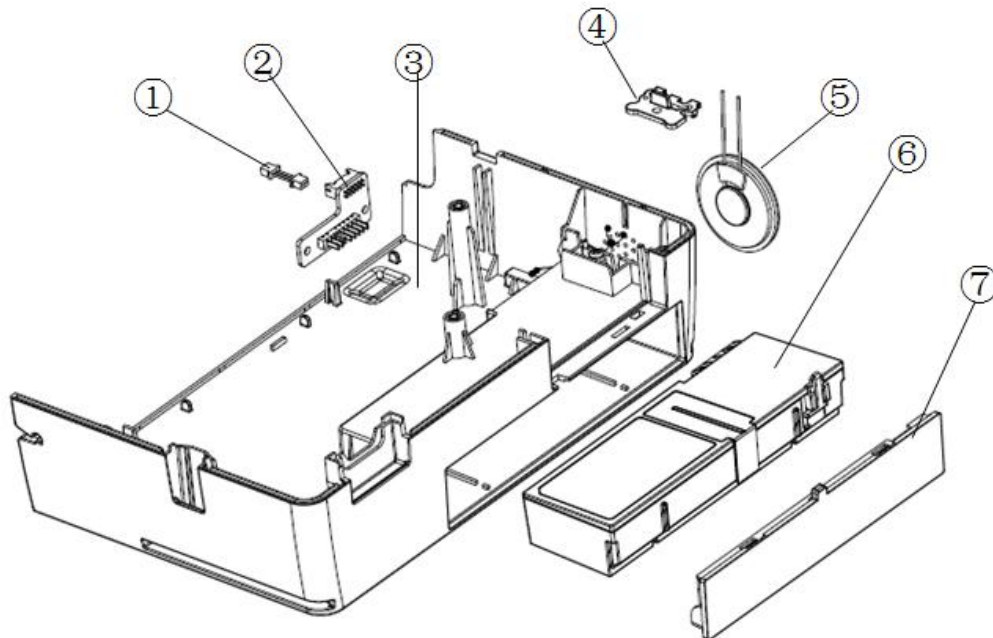


Figure 3 Explosive View of Upper Housing Assembly

6.3.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	009-004120-00	Electric wire 12P double-end connection line, with spacing of 2.0mm and length of 100mm	1
2	/	SP5 battery interface board PCBA (FRU components)	1
3	043-004426-01	VP5 host panel	3
4	043-004209-00	SP5 loudspeaker briquetting	1
5	009-004119-00	Electric wire 3P2, loudspeaker component, with spacing of 1.25mm	1
6	022-000161-00	Li-ion 7.4V 4800mAh SK04B9003	1
7	/	SP5 battery box door (FRU component)	1

Table 2 Parts List of Upper Housing Assembly

6.4 Pump Assembly

6.4.1 Explosive View

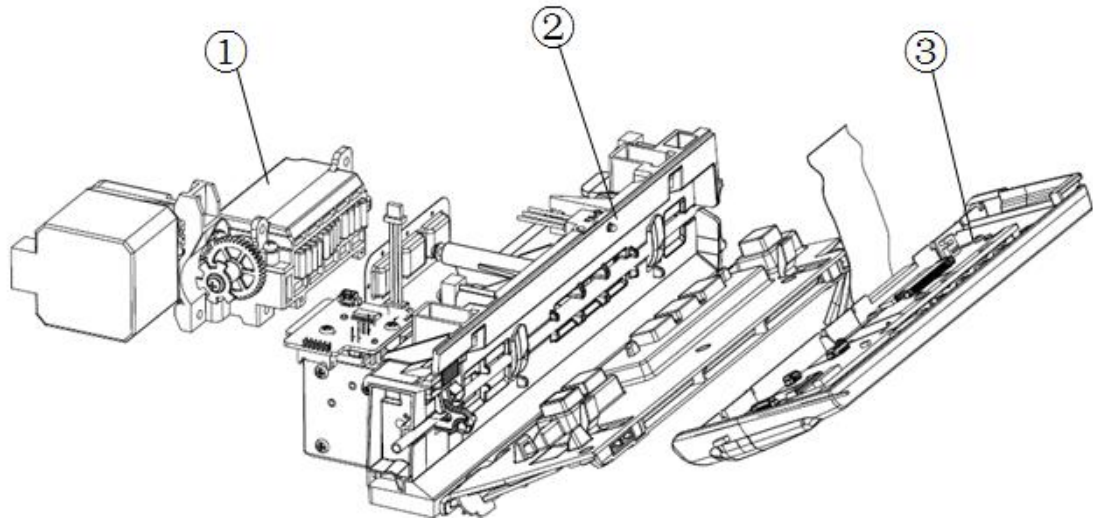


Figure 4 Explosive View of Pump Assembly

6.4.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	115-038471-00	Eccentric gear and motor components	1
2	115-038475-00	Pump body transmission components	1
3	115-038476-00	Pump body transmission components of VP5 ALL infusion pump with specified infusion sets	1
4	115-038472-00	VP5 front door components	1

Table 3 Parts List of Pump Assembly

6.5 Eccentric Gear and Motor Assembly

6.5.1 Explosive View

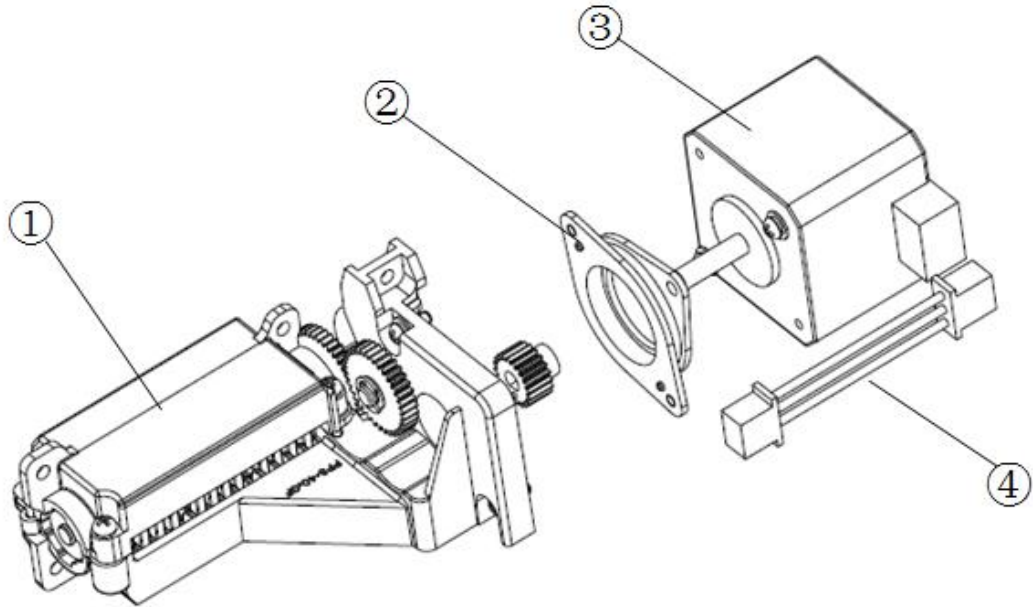


Figure 5 Explosive View of Eccentric Gear and Motor Assembly

6.5.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	/	Eccentric gear components	1
2	034-000069-00	Shock pad	1
3	/	Motor components	1
4	009-006431-00	Motor connection line (4P-6P 80mm)	1

Table 4 Parts List of Eccentric Gear and Motor Assembly

6.6 Pump Transmission Assembly

6.6.1 Explosive View

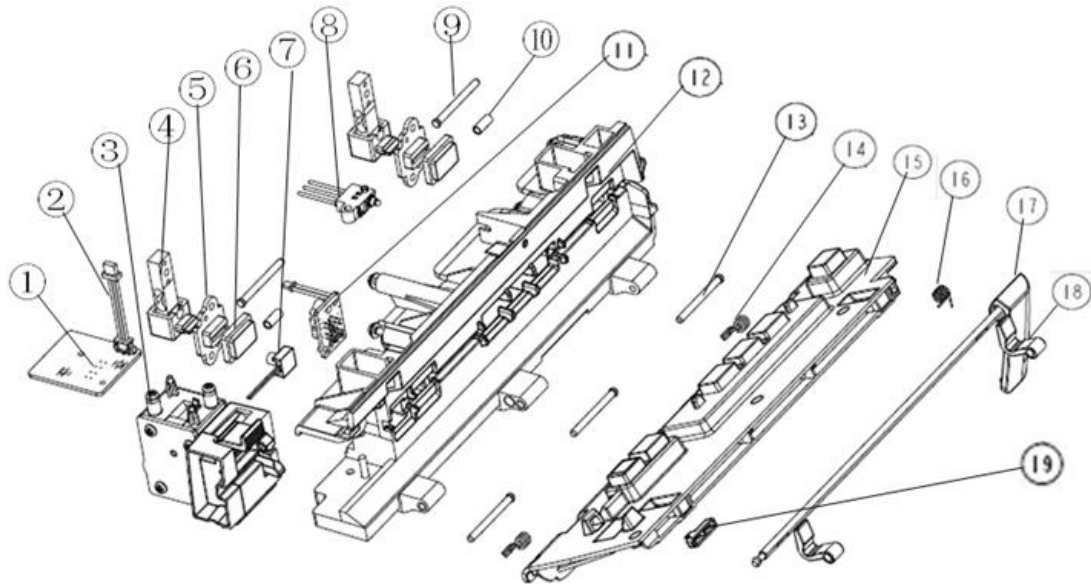


Figure 6 Explosive View of Pump Transmission Assembly

6.6.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	051-002407-00	VP5 ALL downward pressure detection plate PCBA	1
2	009-002717-00	4P Single line	1
3	115-038469-00	VP5 ALL liquid check clip components	1
4	115-036536-00	VP3 drop sensor components	2
5	043-006760-00	Waterproof bracket of pressure sensor	2
6	049-000692-00	VP5 pressure detection silicon waterproof cover	2
7	115-024523-00	VP5 ultrasonic chip components	1
8	009-004534-00	Electric wire 3P ALPS position switch wire, with spacing of 2.54mm	1
9	041-013053-00	VP5 door pivot	2
10	041-016597-00	VP5 door spindle lantern ring	2
11	051-002406-00	VP5 ALL motor detection plate PCBA	1
12	043-007025-00	VP5 pump body mounting base (VP5 ALL- conventional infusion sets)	1
12	043-007026-00	VP5 pump mounting base (VP5 ALL- specified infusion sets)	1
13	041-013053-00	VP5 door pivot	3
14	033-000331-00	VP5 door buffering torsion spring (ID3.5x6)	2
15	043-007028-00	Rear housing of door of VP5 ALL	1

15	043-007461-00	VP5 ALL door rear housing (specified infusion sets)	1
16	033-000329-00	VP5 door holder torsion spring (ID4.5x6)	1
17	115-024514-00	VP5 door handle components	1
18	044-000713-00	VP5 door grab metal mould	2
19	043-005154-00	VP5 stop block of door pull rod	1

Table 5 Parts List of Pump Transmission Assembly

6.7 VP5 Front Door Housing Assembly

6.7.1 Explosive View

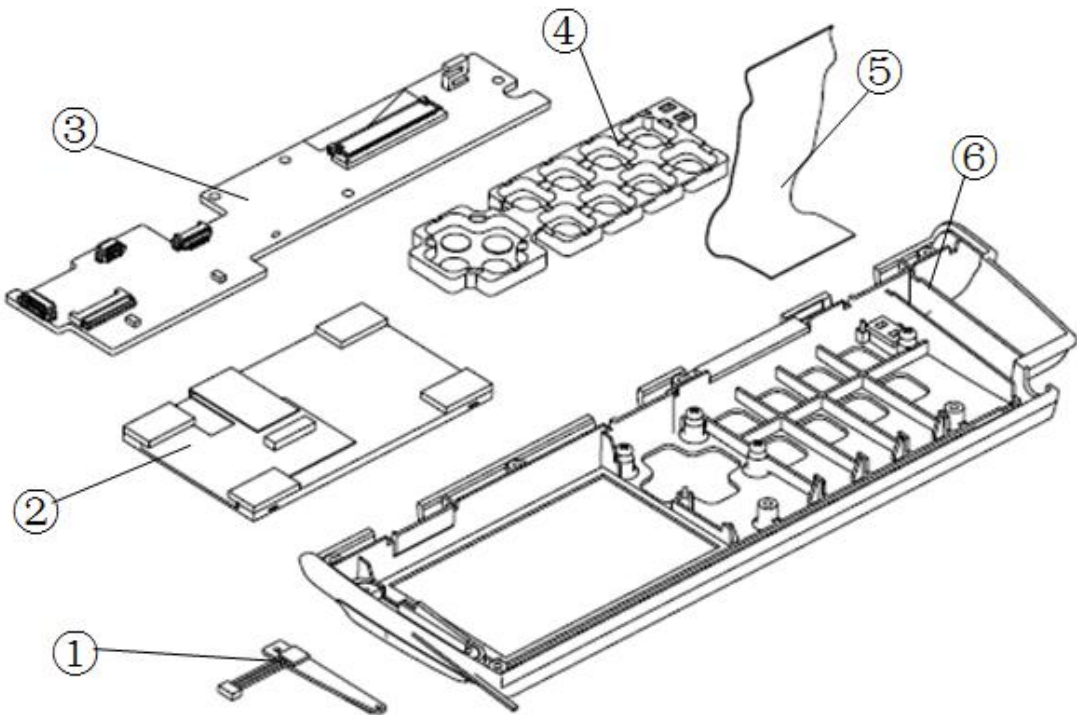


Figure 7 Explosive View of VP5 Front Door Housing Assembly

6.7.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	/	VP5 alarm light (FRU components)	1
2	/	VP5&SP5 display	1
3	/	VP5 keyboard (FRU components)	4
4	049-001059-00	VP5&SP5 silicon keys	1
5	009-004212-01	Wire FFC, electric wire 30P, with spacing of 1.0mm, different interface	1
6	043-007028-00	Front housing of door	1

Table 6 Parts List of VP5 Front Door Housing Assembly

6.8 VP5 Housing Base Assembly

6.8.1 Explosive View

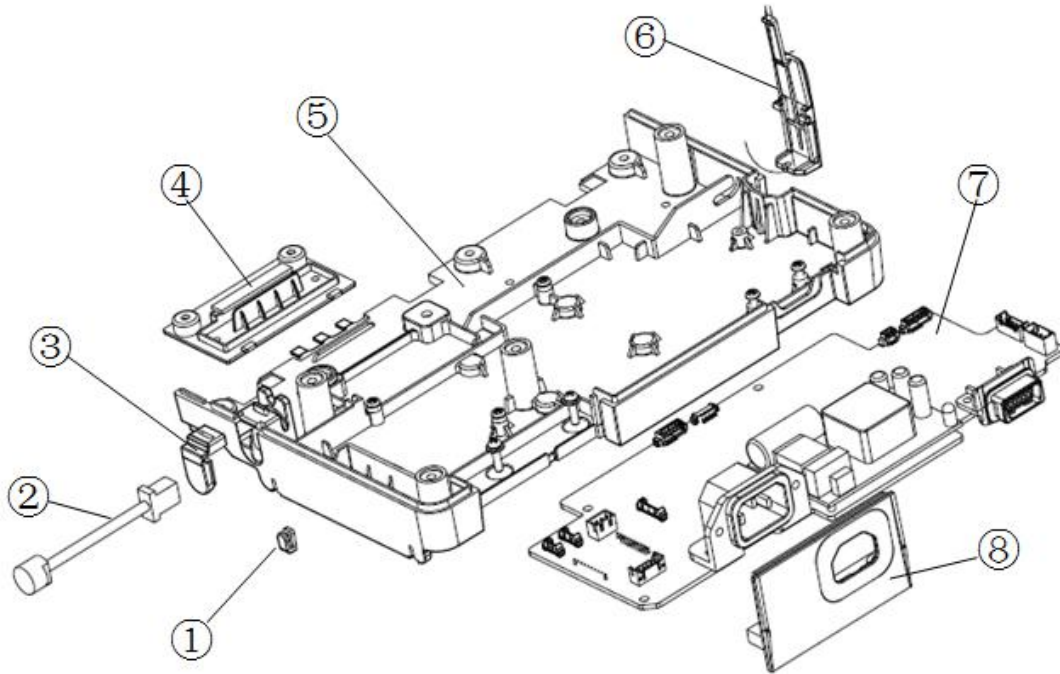


Figure 8 Explosive View of VP5 Housing Base Assembly

6.8.2 Parts List

Serial No.	Sub-part code	Sub-part description	Quantity
1	049-000664-00	Floor pad	4
2	008-000841-00	Drop sensor socket	1
3	049-000661-00	VP5 drop interface rubber plug	1
4	043-004406-00	VP5 bottom housing plug wire cap	1
5	043-004428-02	VP5 bottom housing	1
6	043-004208-00	VP5&SP5 stacking hasp (common mould)	1
7	051-002405-00	VP5 main control board PCBA	1
7	051-002582-00	VP5 ALL main control board PCBA (with drop circuit)	1
8	043-004215-01	AC socket support (after encapsulation)	1

Table 7 Parts List of VP5 Housing Base Assembly

7 Upgrading

7.1 Tools

During upgrading, you may need the following tools:

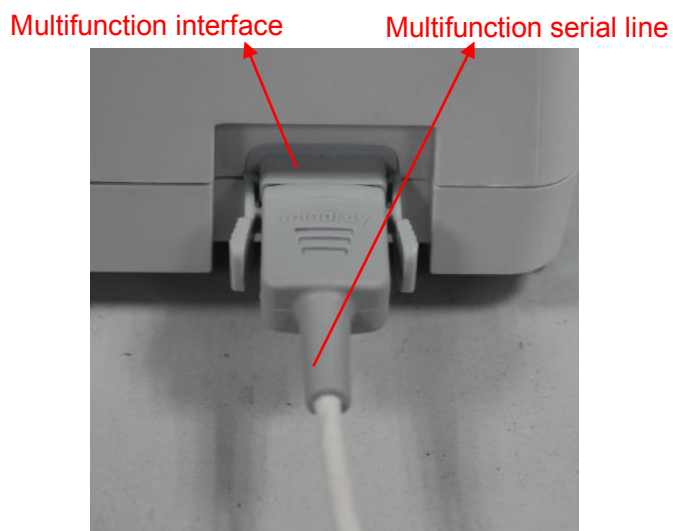
- ◆ Multi-functional serial line
- ◆ PC
- ◆ PC burning tool software

7.2 Upgrading

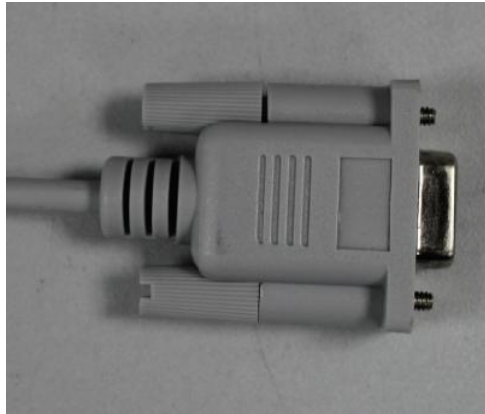
This infusion pump supports software upgrade. The burning software can be directly run on a desktop PC or mobile PC. Through a multifunction interface, the infusion pump can be connected to a PC to upgrade software and drug library, update built-in infusion set brands, view and export history records and abnormal records.

7.2.1 Software Burning Method

1. Insert the multifunction serial line to the multifunction interface on the host as shown in the figure.



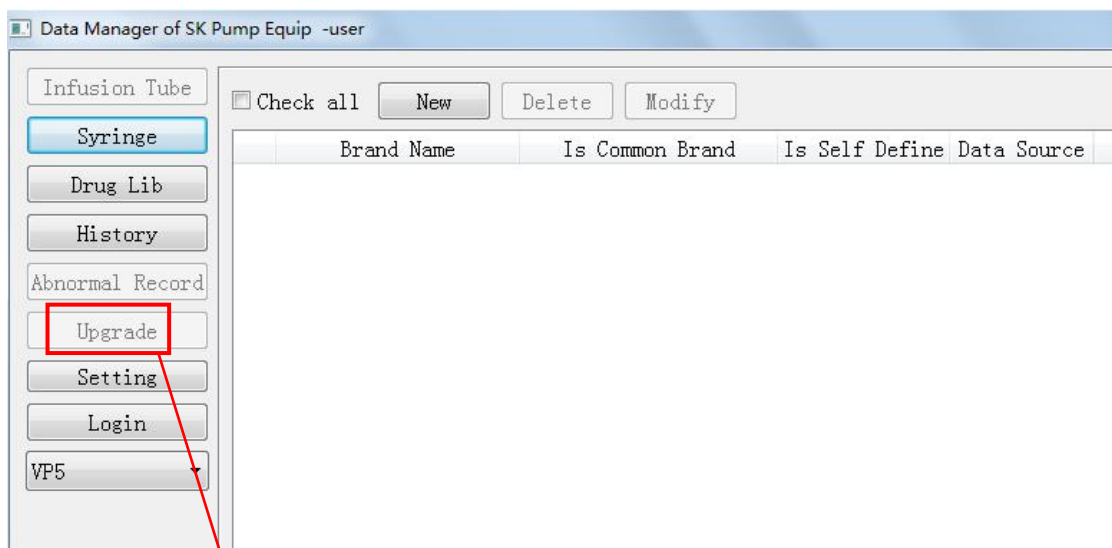
2. The other end of the multifunction serial line is a standard DB9 serial port. Connect this end to PC as shown in the figure.



3. Press  and  simultaneously, and the machine will enter the "BootLoader" interface as shown in the figure.



4. Open the PC burning tool as shown in the figure. At this time the "Upgrade" menu is not available and may not be selected.

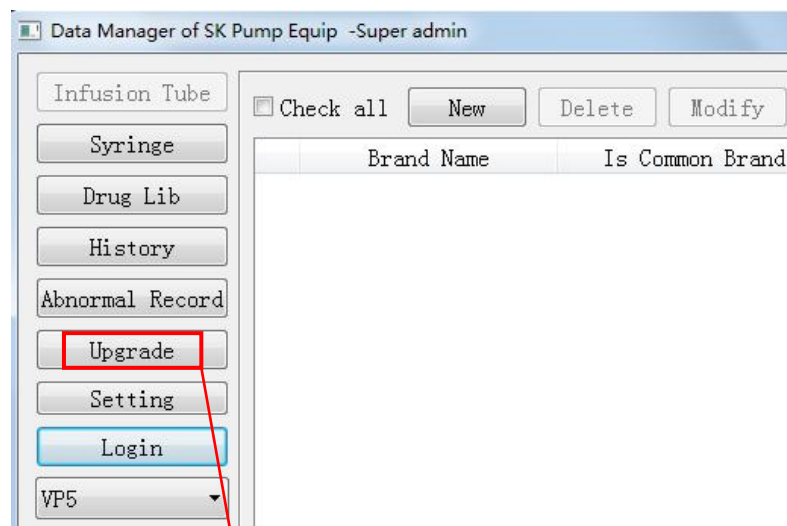


"Upgrade" menu may not be used at this time.

5. Click " Login" and login password dialog box pops up, as shown in the figure.

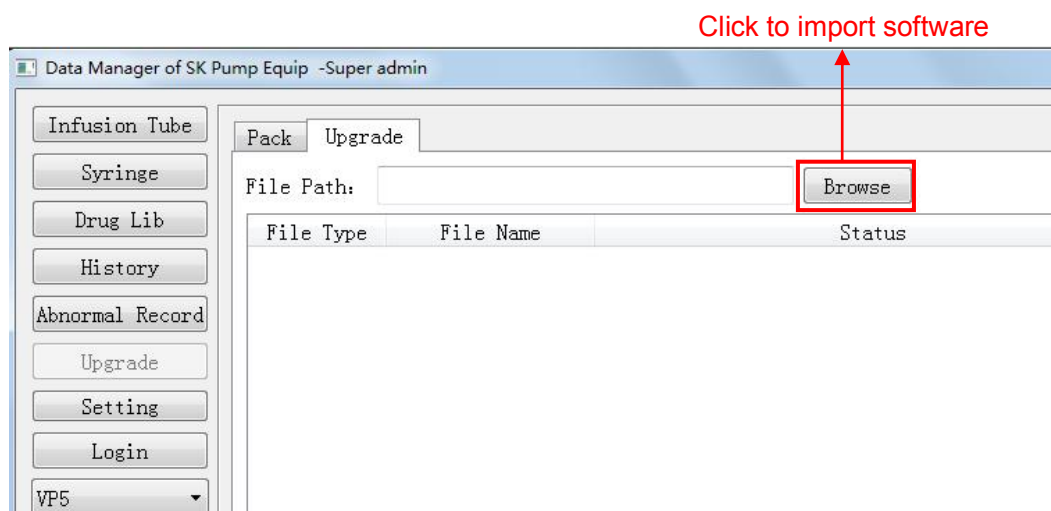


6. Input password and now upgrade menu becomes available, as shown in the figure.

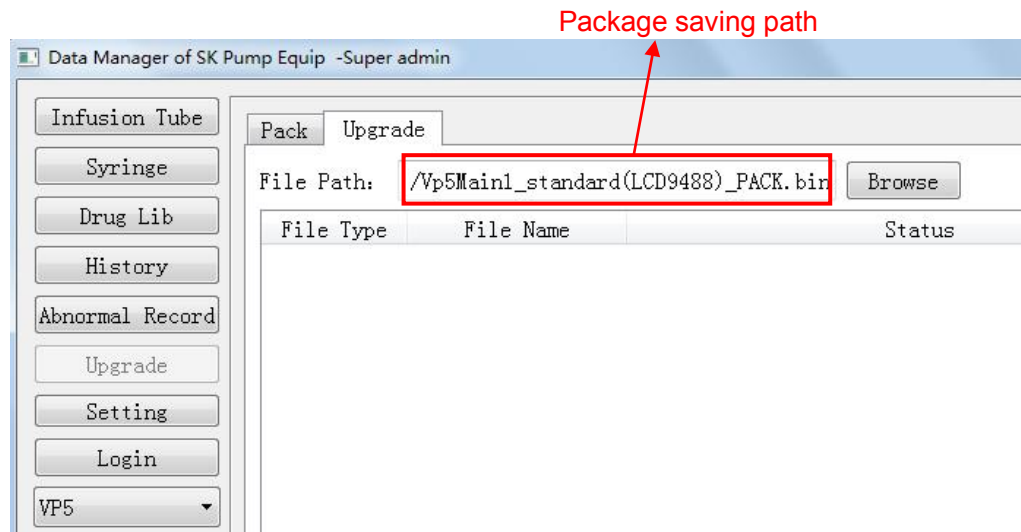


"Upgrade" menu is available now.

7. Select "Upgrade" and you will enter the interface as shown in the figure:

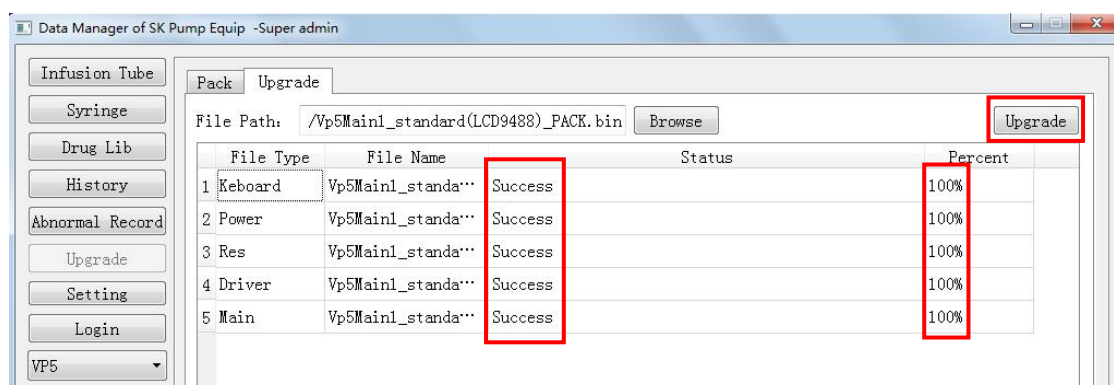


8. In the upgrade interface, select "Browse" and choose the software package to be burnt as shown in the figure.



9. Click "Upgrade" at the upper right corner of the PC burning tool software and the software starts burning.

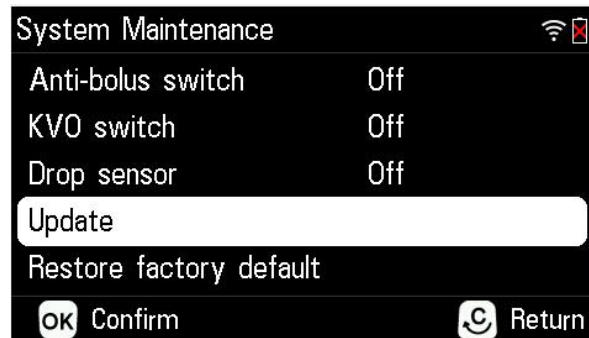
10. After the burning is completed, upgrade status indicates "Success" and upgrade percent indicates "100%" as shown in the figure.



11. Turn off the pump and disconnect the multifunction serial line. Now, the software burning is completed.

7.2.2 Update Method of Drug Library

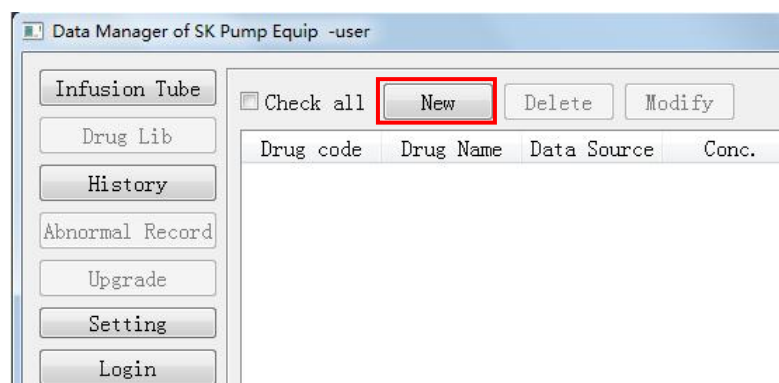
1. Turn on the pump, Select [Main Menu]→[System Maintenance].
2. Input password “6686”, press  to confirm.
3. Select [Update] by pressing  or .



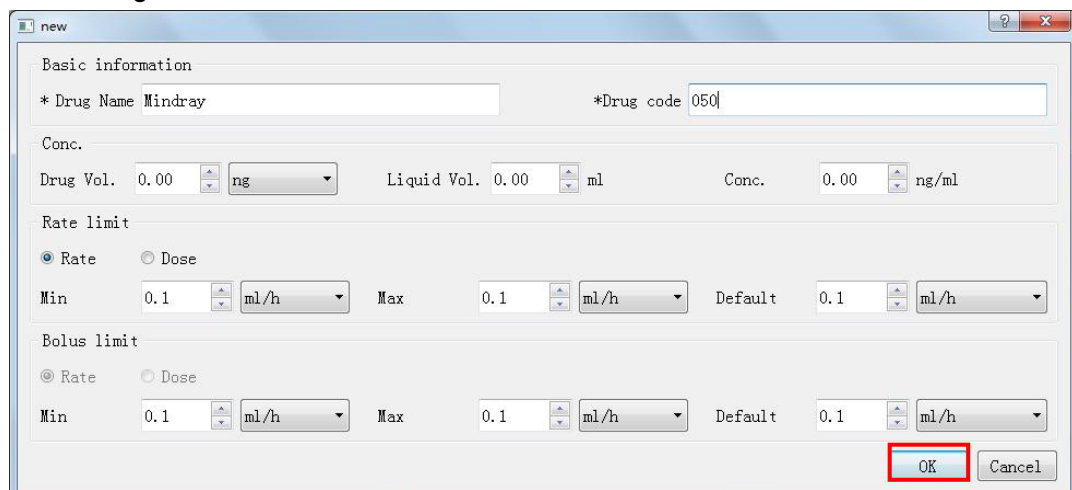
4. Connect the multifunction serial line to PC.

7.2.2.1 Add of Single Drug Library

1. Open the PC burning tool, select “Drug Lib” and click “New”.

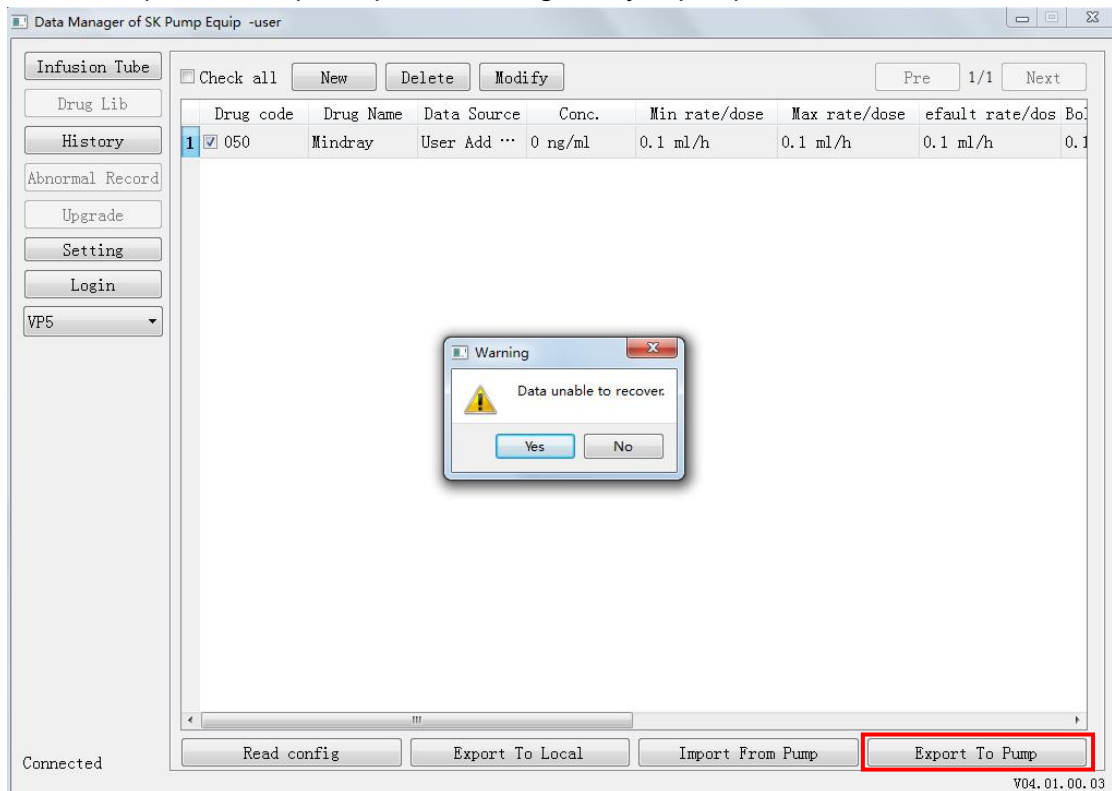


2. Edit drug information, and click “OK”.

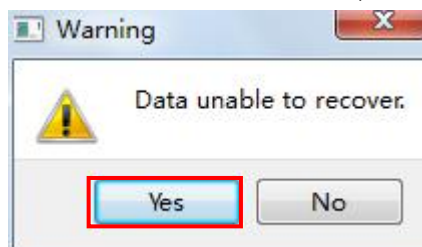


3. Check the newly added drug name.

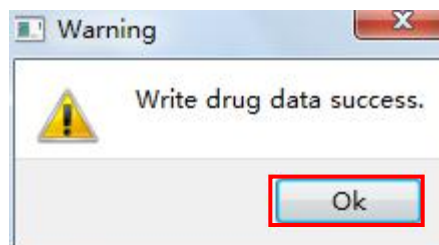
4. Click “Export To Pump” to update the drug library to pump.



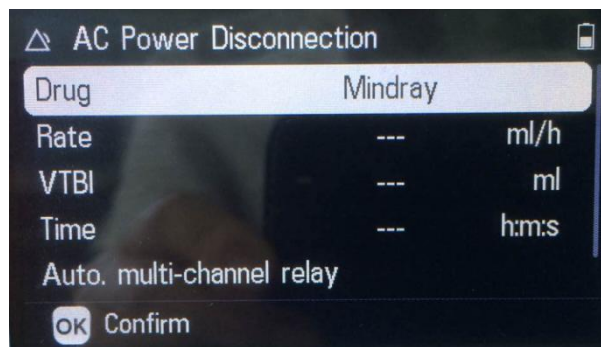
5. You will be prompted with “Data unable to recover”, click “Yes”.



6. Then you will be prompted with “Write drug data success.”, click “OK” to confirm reboot the pump.



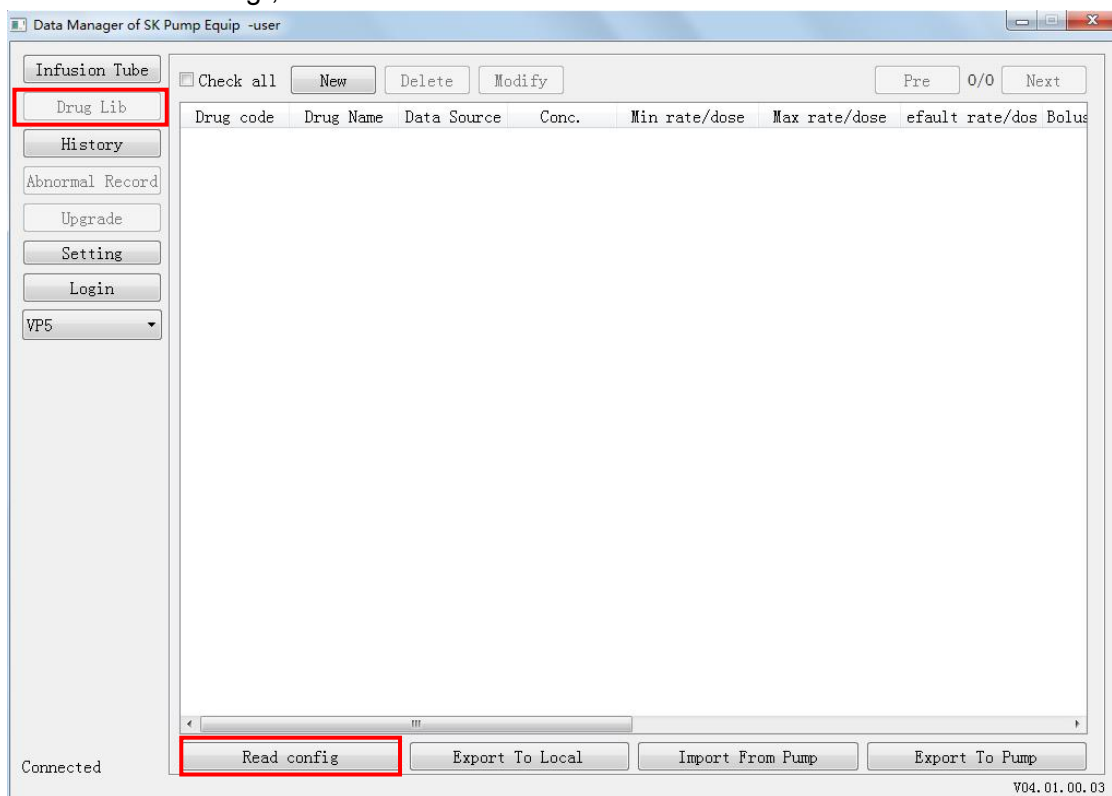
-
7. Confirm whether the drug library is added to the pump.



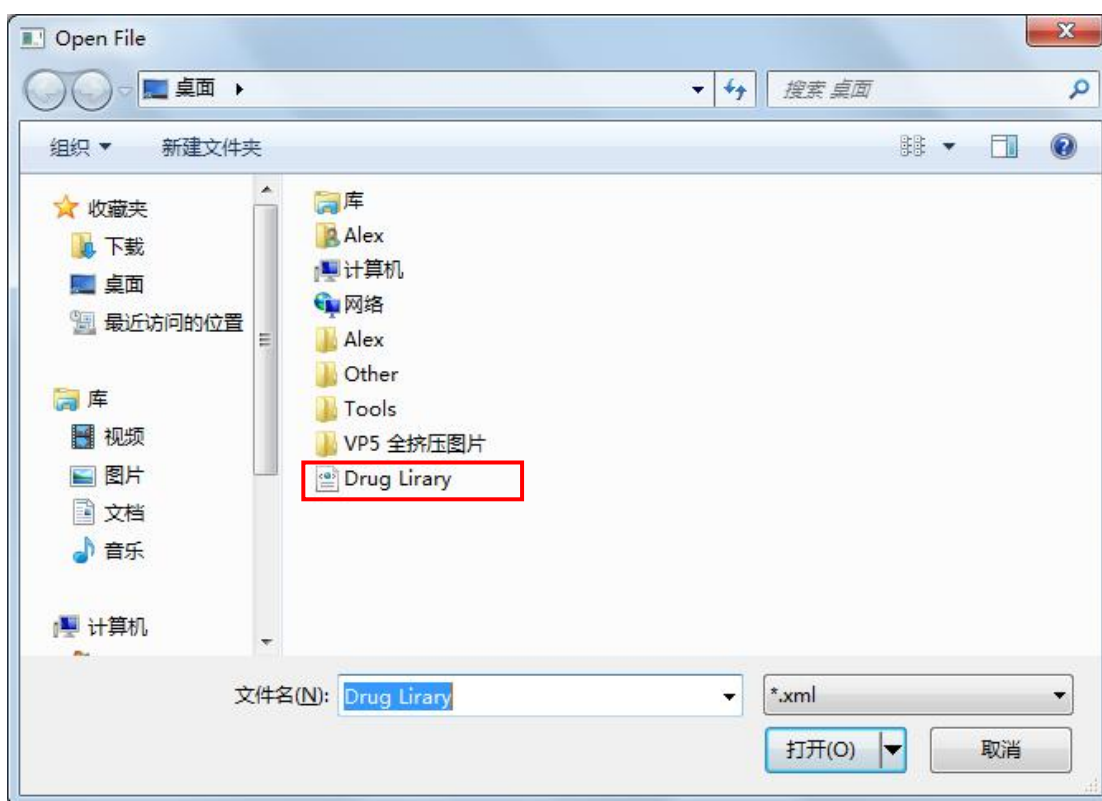
7.2.2.2 Import of Batch Drug Library

1. Source file (xml format) of drug library is required when importing batch drug library.

2. Click “Read config”, and find the source file in PC.

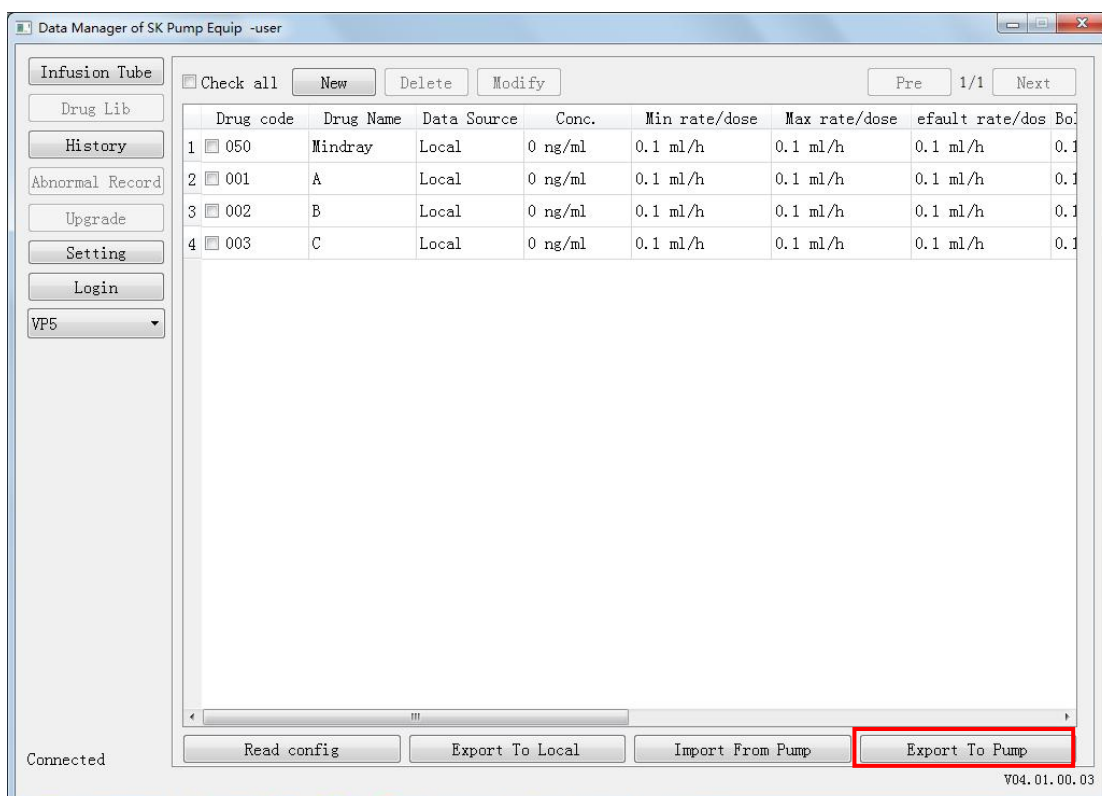


3. Click “.xml” file as shown in the figure.

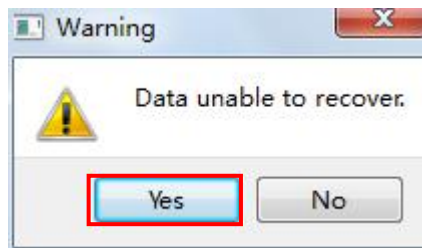


4. Drugs in source file of drug library shall be displayed as shown in the figure.

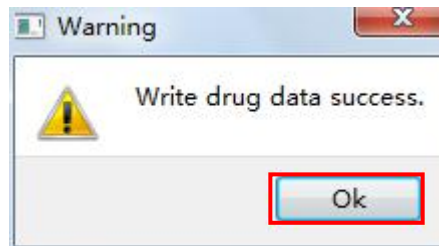
5. Check the drug name, click “Export To Pump” to update the drug library to pump.



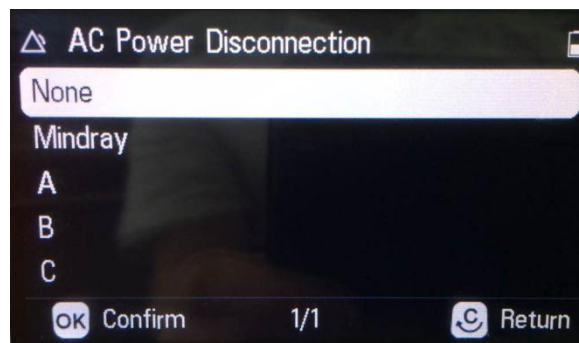
6. You will be prompted with “Data unable to recover”, click “Yes”.



7. Then you will be prompted with “Write drug data success.”, click “OK” to confirm reboot the pump.



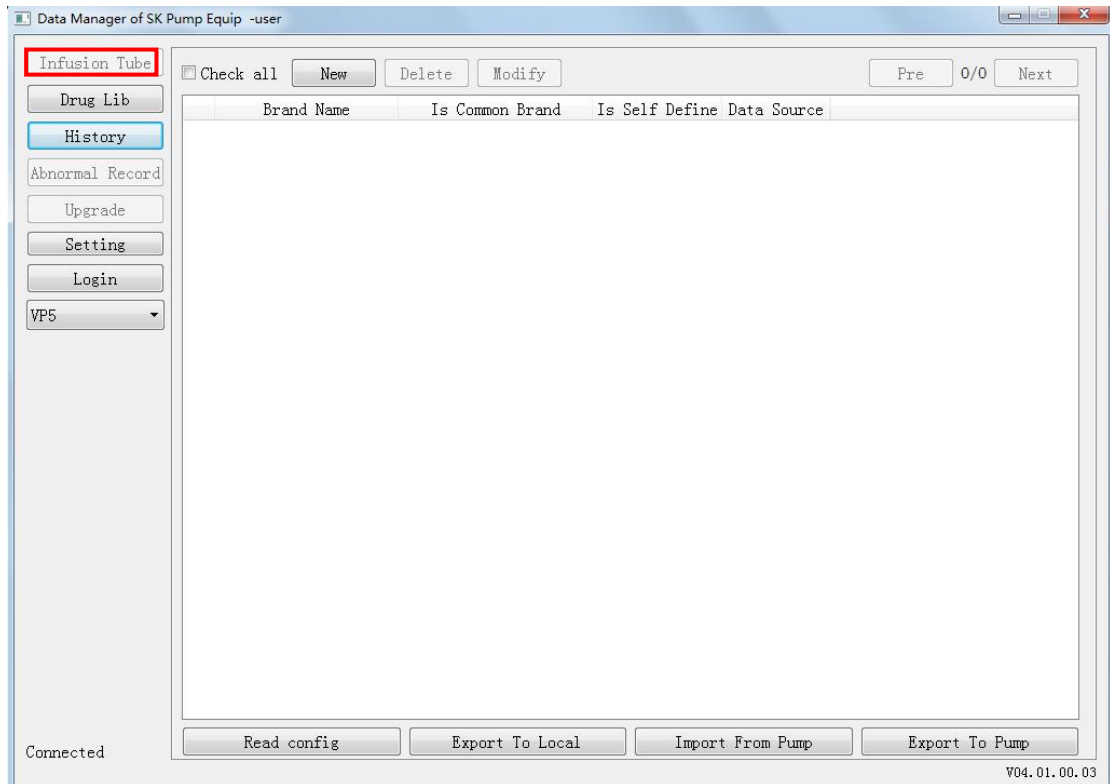
8. Confirm whether the drug library is added to the pump.



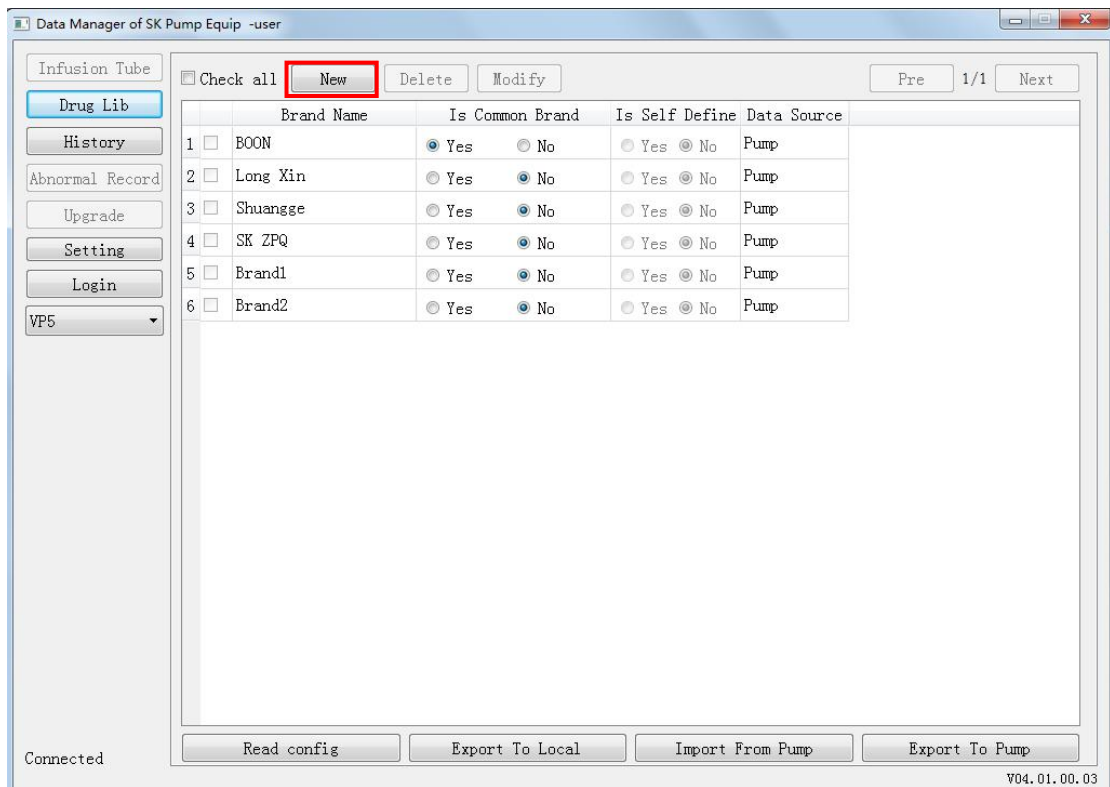
Note: Click “Export To Local” to save the checked drug library of PC burning tool to PC.

7.2.3 Add Infusion Tube Brand to BeneFusion VP5

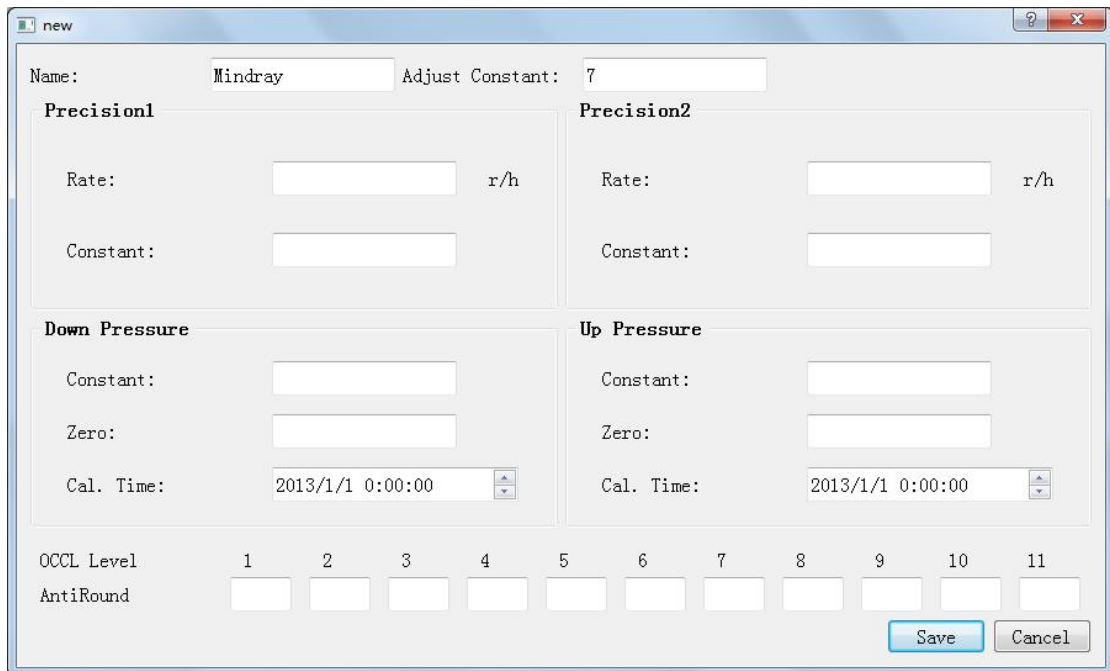
1. Connect the multifunction serial line of VP5 to PC (Please refer to **7.2.2 Update Method of Drug Library** for detailed connection method).
2. Click “Infusion Tube” after successful connection.



3. PC burning tool will automatically read the data of infusion tube brand of VP5.
4. Click “New” to add other brands of infusion tube to VP5.

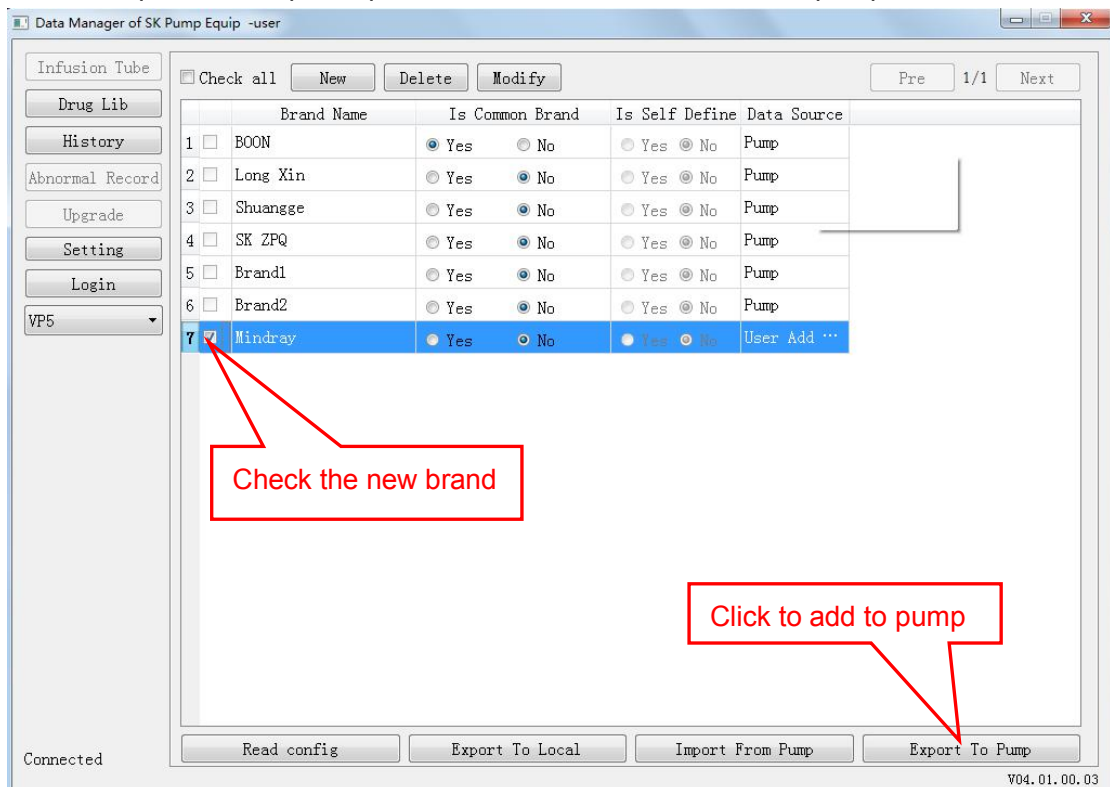


5. Edit the name and parameters of newly added infusion tube (Name and specification are required).

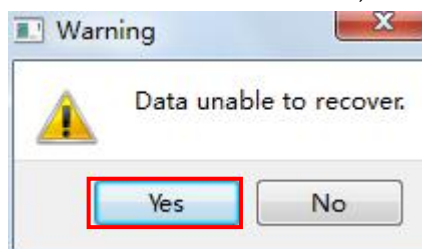


6. Check the new brand of infusion tube.

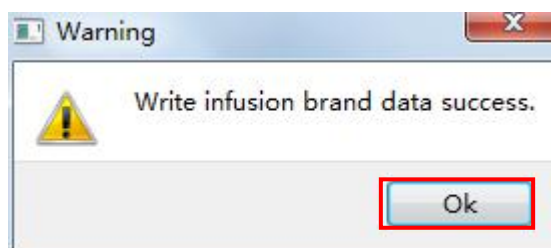
7. Click “Export To Pump” to update the brands of infusion tube to pump.



8. You will be prompted with “Data unable to recover”, click “Yes”.



9. PC burning tool will display the process of updating the brands of infusion tube, you will be prompted with “Write infusion brand data success.”, click “OK” to confirm reboot the pump.



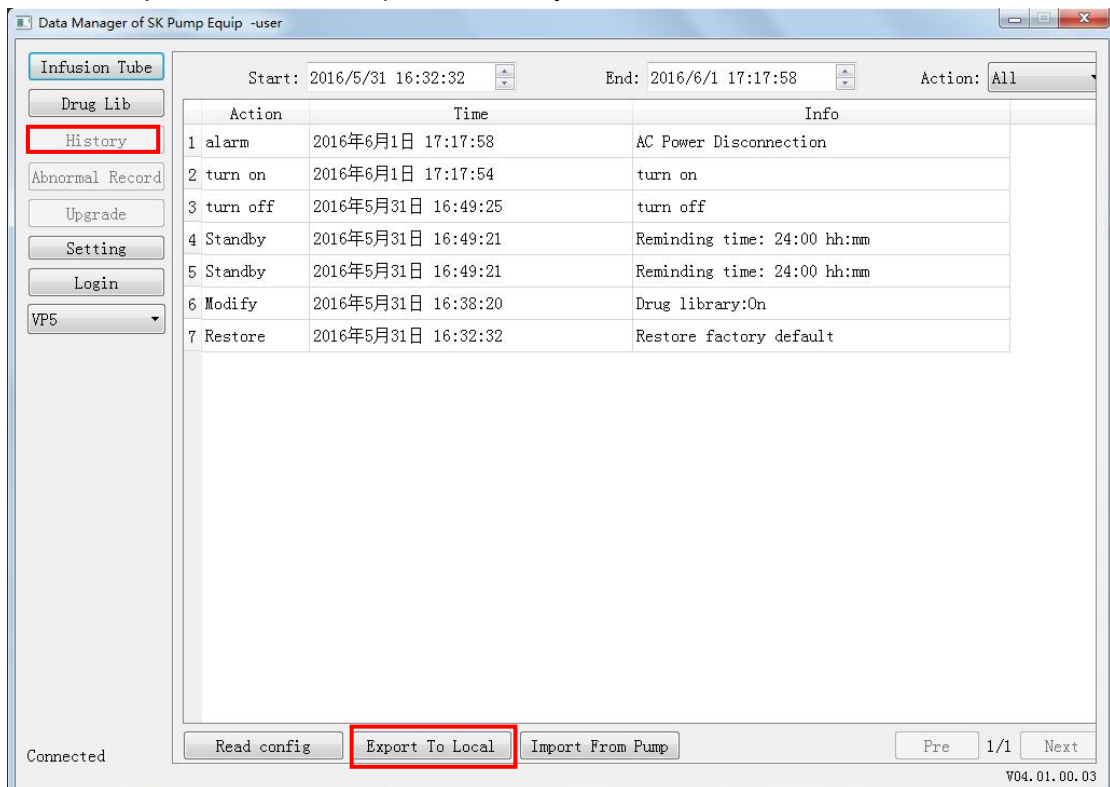
10. Reboot the pump to confirm whether “Mindray” brand is added to the pump.

Note 1: Click “Export To Local” to save the checked brand and parameters of infusion tube of PC burning tool to PC.

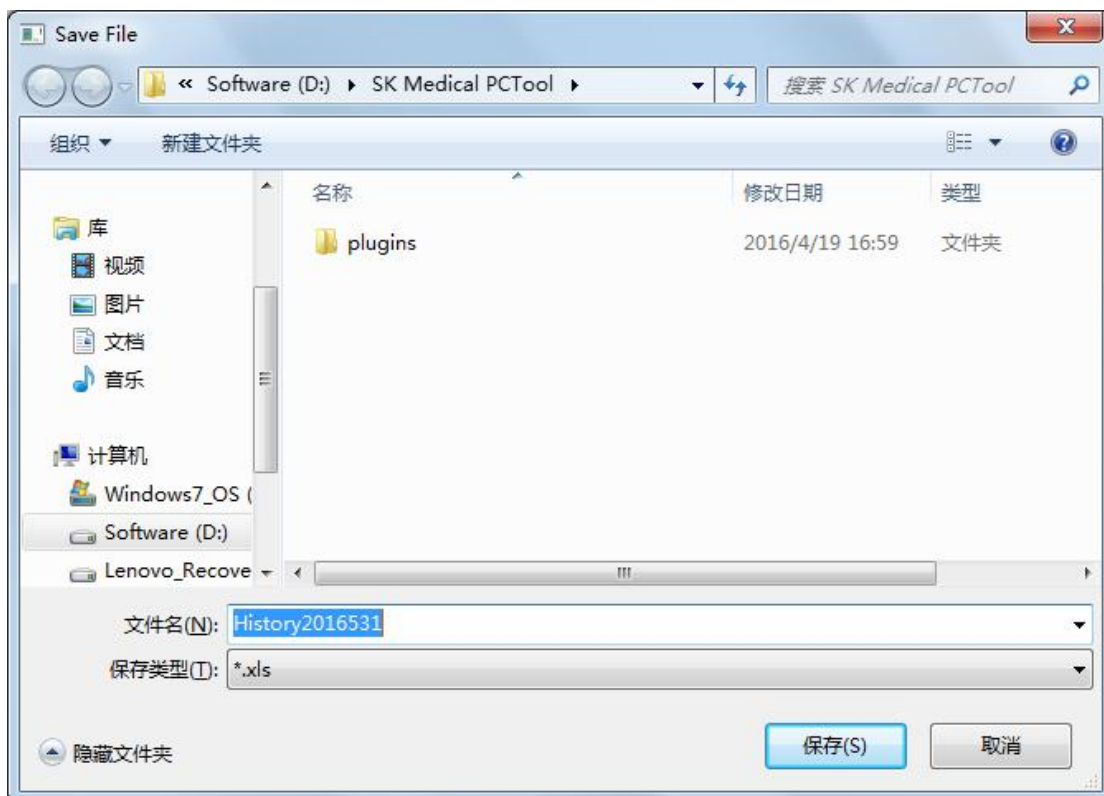
Note 2: Please perform accuracy calibration and pressure calibration before adding infusion tube brand, to guarantee the accuracy of newly added infusion tube.

7.2.4 Export History Records to PC

1. Connect the multifunction serial line of VP5 to PC (Please refer to **7.2.2 Update Method of Drug Library** for detailed connection method).
2. Click “History” after successful connection, PC burning tool will automatically read the history records of VP5.
3. Click “Export To Local” to export the history records.



4. Select the saving path of history records.
5. Edit the file name and click “Save”.



6. Double click the saved history record after operation, so as to check whether the history record is saved successfully.

7.2.5 Export Abnormal Records to PC

The methods of exporting “Abnormal records” is the same as that of “History record”, please refer to **7.2.4 Export History Records to PC** for details.

